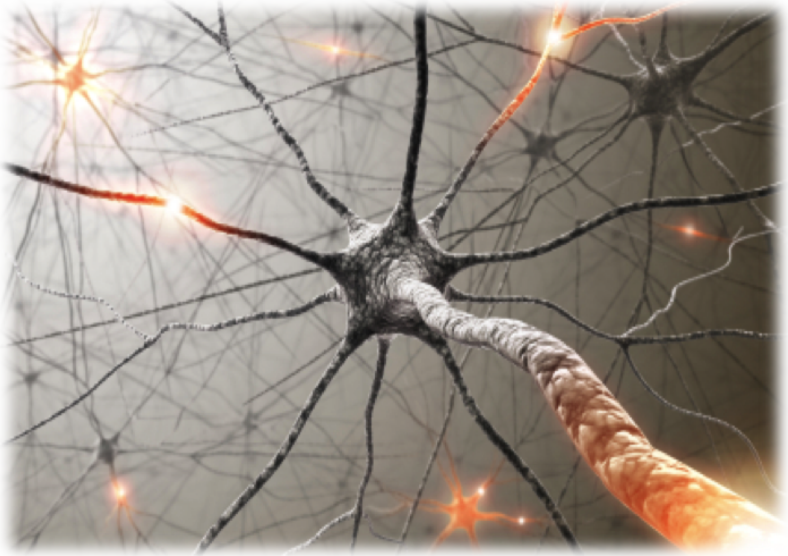
Lecture 3: Machine Learning 2



Roadmap

Group DRO

Non-linear features (offline)

Feature templates (offline)

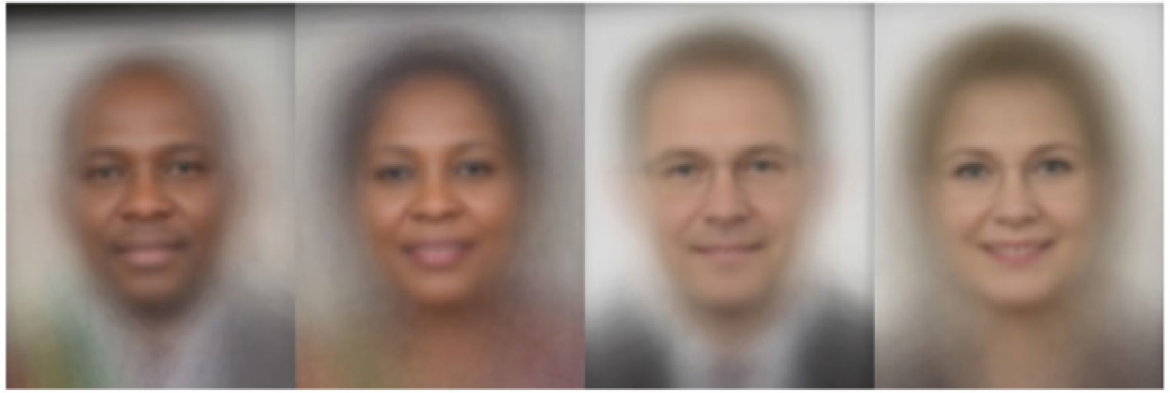
Neural networks

Backpropagation

- Thus far, we have focused on finding predictors that minimize the training loss, which is an average (of the loss) over the training examples.
- While averaging seems reasonable, in this module, I'll show that averaging can be problematic and lead to inequalities in accuracy across groups.
- Then I'll briefly present an approach called **group distributional robust optimization (group DRO)**, which can mitigate some of these inequalities.

Gender Shades

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%



Inequalities arise in machine learning

- is the Gender Shades project by Joy Buolamwini and Timnit Gebru (paper).
- In this project, they collected an evaluation dataset of face images, which aimed to have balanced representation of both faces with lighter and darker skin tones, and across men and women. To do this they curated the public face images from members of parliament across several African and European countries.
- Then, they evaluated three commercial systems, from Microsoft, Face++, and IBM, on the task of gender classification (which in itself maybe a dubious task, but was one of the services offered by these companies).
- The results were striking: all three systems got nearly 100% accuracy for lighter-skinned males, but they all got much worse accuracy for darker-skinned females.
- Gender Shades is just one of many examples pointing at a general and systemic problem: machine learning models, which are typically optimized to maximize average accuracy, can yield poor accuracies for certain **groups** (subpopulations).
- These groups can be defined by protected classes as given by discrimination law such as race and gender, or perhaps defined by the user identity or location.

False arrest due to facial recognition

Real-life consequences

- Poor accuracy of machine learning systems can have serious real-life consequences. In one vivid case, a Black man by the name of Robert Julian-Borchak Williams was wrongly arrested due to a incorrect match with another Black man captured from a surveillance video, and this mistake was made by a facial recognition system.
- Given the Gender Shades project, we can see that lower accuracies for some groups might even lead to more arrests, which adds to the already problematic inequalities that exist in our society today.
- In this module, we'll focus only on performance disparities in machine learning and how we can mitigate them.
- But even if facial recognition worked equally well for all groups of people, should it even be used for law enforcement? Should it be used at all? These are bigger ethical questions, and it's always important to remember that sometimes, the issue is not with the solution, but the framing of the problem itself.

Linear regression with groups

training data

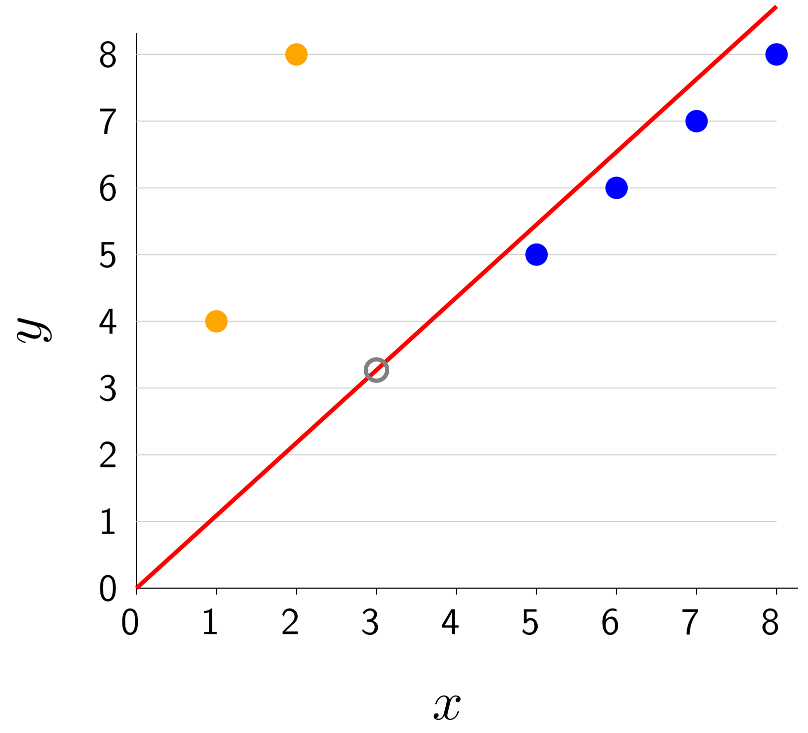
	x	y	g
example	1	4	A
example	2	8	A
example	5	5	B
	6	6	B
	7	7	B
	8	8	B

group

learning algorithm → f predictor

3 input

3.27 output



$$f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x) \qquad \mathbf{w} = [w] \qquad \phi(x) = [x]$$

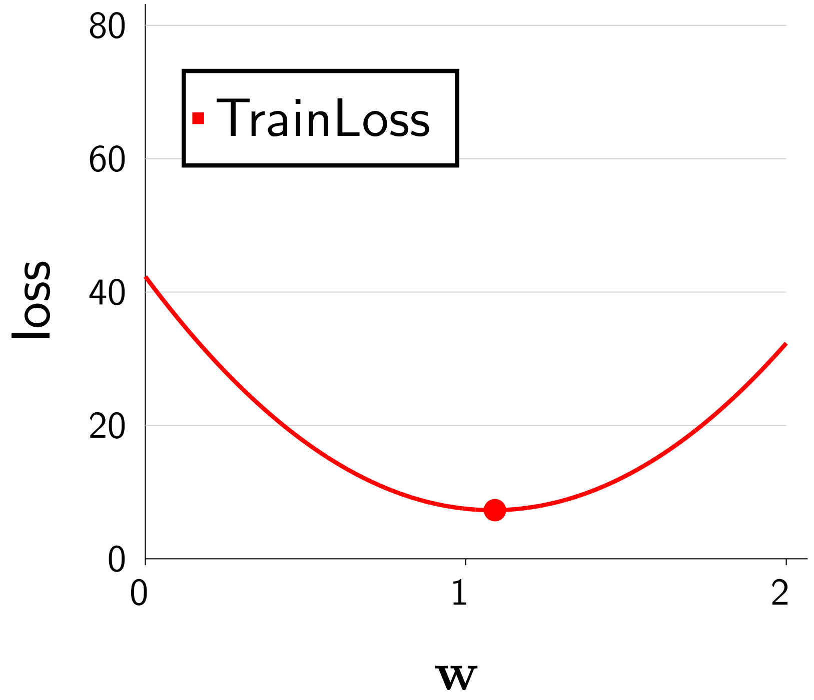
Note: predictor $f_{\mathbf{w}}$ does not use group information g

- Gender Shades was an example of classification, but to make things simpler, let us consider linear regression.
- We start with training data, but this time, each example consists of not only the input x and the output y, but also a **group** g (e.g., gender).
- In this example, we will assume we have two groups, A and B. As we see on the plot, the A points rise quickly, and the B points lie on the identity line, so the points behave differently.
- Recall the goal of regression is to produce a predictor that can take in a new input and predict an output.
- In linear regression, each predictor $f_{\mathbf{w}}$ computes the dot product of the weight vector $\mathbf{w}$ and a feature vector $\phi(x)$, and for this example, we define the feature map $\phi(x)$ to be the identity map, so that our hypothesis class consists of lines that pass through the origin.
- Looking ahead a bit, we see that there is a bit of a tension, where what slope w is best for group A is not the same as what is best for group B. The question is how we can compromise.
- Note that in this setting, the predictor $f_{\mathbf{w}}$ does not use the group information g, so it cannot explicitly specialize to the different groups. The group information is only used by the learning algorithm as well as to evaluate the performance across different groups.

Average loss

$$\text{Loss}(x, y, \mathbf{w}) = (f_{\mathbf{w}}(x) - y)^2$$

x	y	g
1	4	A
2	8	A
5	5	B
6	6	B
7	7	B
8	8	B



$$\text{TrainLoss}(\mathbf{w}) = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{(x,y) \in \mathcal{D}_{\text{train}}} \text{Loss}(x, y, \mathbf{w})$$

$\text{TrainLoss}(1) = \frac{1}{6}((1-4)^2 + (2-8)^2 + (5-5)^2 + (6-6)^2 + (7-7)^2 + (8-8)^2) = 7.5$

- Recall that in regression, we typically use the squared loss, which measures how far away the prediction $f_{\mathbf{w}}(x)$ is away from the target y.
- Also recall that we defined the training loss to be an average of the per-example losses. This gives us a loss value for each value of $\mathbf{w}$ (see plot).
- So if we evaluate the training loss at $\mathbf{w} = 1$, we're averaging over the 6 examples. For each example, the prediction $f_{\mathbf{w}}(x)$ is just x (since $\mathbf{w} \cdot \phi(x) = [1] \cdot [x] = x$, so we can average the squared differences between x and y, producing a final answer of 7.5.
- If you evaluate $\mathbf{w}$ at a different value, you get a different training loss.
- If we minimize the training loss, then we can find this point $\mathbf{w} = 1.09$.

Per-group loss

x	y	g
1	4	A
2	8	A
5	5	B
6	6	B
7	7	B
8	8	B

$$\text{TrainLoss}_g(\mathbf{w}) = \frac{1}{|\mathcal{D}_{\text{train}}(g)|} \sum_{(x,y)\in\mathcal{D}_{\text{train}}(g)} \text{Loss}(x, y, \mathbf{w})$$

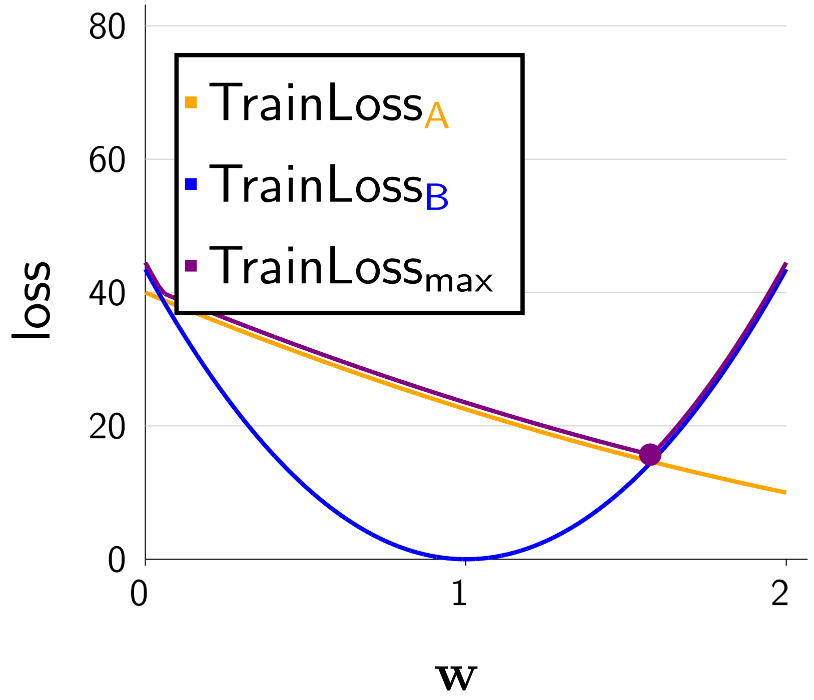
$$\text{TrainLoss}_{\text{A}}(1) = \tfrac{1}{2}((1-4)^2 + (2-8)^2) = 22.5$$

$$\text{TrainLoss}_{\text{B}}(1) = \tfrac{1}{4}((5-5)^2 + (6-6)^2 + (7-7)^2 + (8-8)^2) = 0$$

- Let us now take a careful look at the loss of each group.
- First, define $\mathcal{D}_{\text{train}}(g)$ to be the set of examples in group g. For example, $\mathcal{D}_{\text{train}}(\text{A}) = \{(1, 4), (2, 8)\}$.
- Then define the **per-group loss** TrainLoss_g of a weight vector $\mathbf{w}$ to be the average loss over the points in group g.
- If we choose $\mathbf{w} = [1]$ as our running example (because it's easy to compute), we see that the per-group loss on group A is 22.5, while the per-group loss on group B is 0.
- So note that even though the average loss was only 7.5, there is a huge performance disparity, with group A suffering a much larger loss.

Maximum group loss

x	y	g
1	4	A
2	8	A
5	5	B
6	6	B
7	7	B
8	8	B



$$\text{TrainLoss}_{\text{max}}(\mathbf{w}) = \max_g \text{TrainLoss}_g(\mathbf{w})$$

$\text{TrainLoss}_{\text{A}}(1) = 22.5$

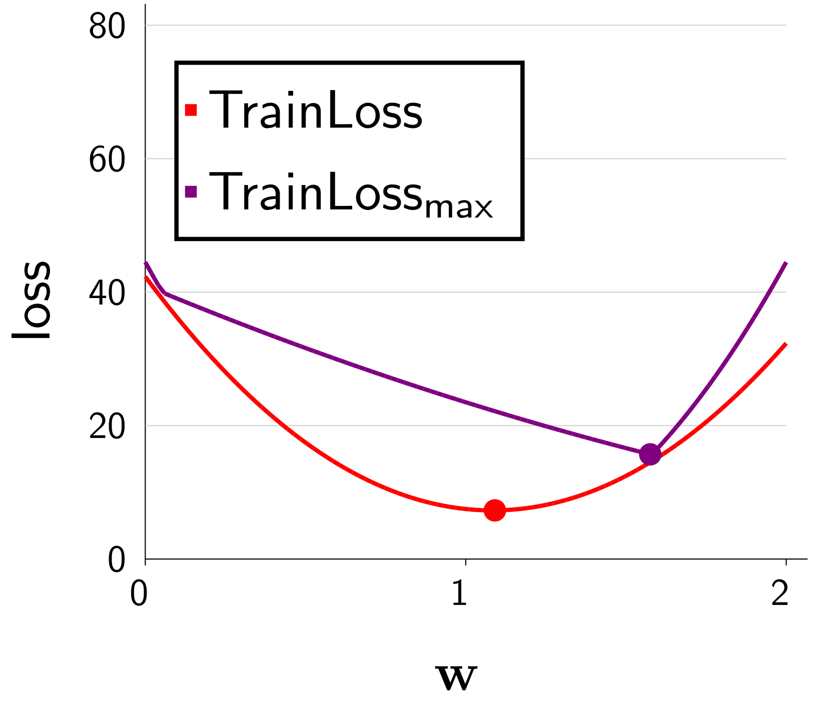
$\text{TrainLoss}_{\text{B}}(1) = 0$

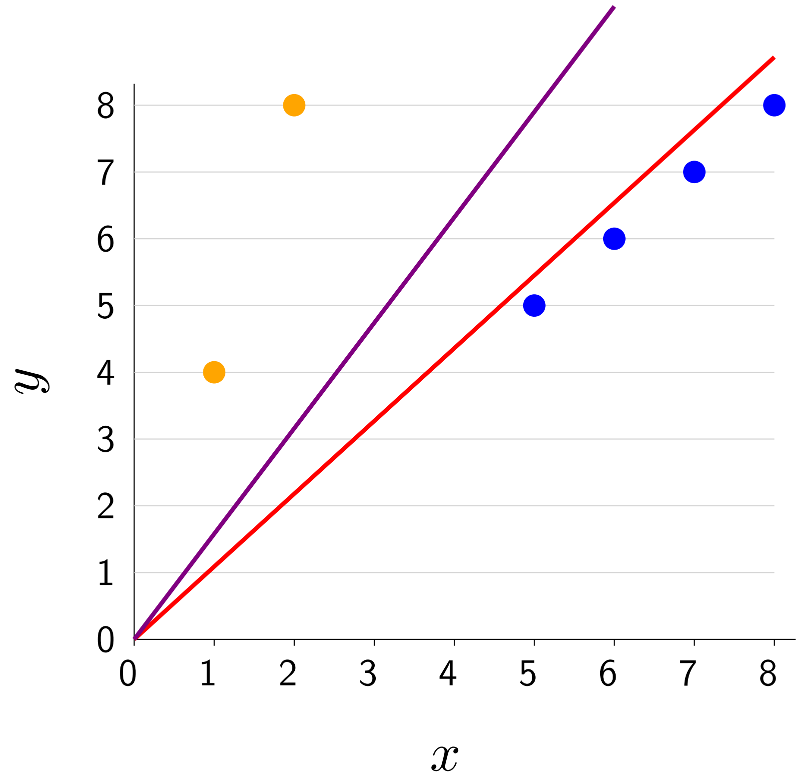
$\text{TrainLoss}_{\text{max}}(1) = \max(22.5, 0) = 22.5$

- We now want to somehow capture the per-group losses by one number. To do this, we look at the worst-case over groups.
- We now define the **maximum group loss** to be the largest over all groups. This function is the pointwise maximum of the per-group losses, which you can see on the plot as taking the upper envelope of the two per-group losses.
- We call this method group distributionally robust optimization (group DRO), because it is a special case of a broader framework Distributionally robust optimization (DRO).
- Going back to our running example with $\mathbf{w} = [1]$, we take the maximum of 22.5 and 0, which is 22.5.
- Note that this is much higher than the average loss (7.5), signaling that some group(s) are far less well off than the average.

Average loss versus maximum group loss

x	y	g
1	4	A
2	8	A
5	5	B
6	6	B
7	7	B
8	8	B





Standard learning:

minimizer of average loss: $\mathbf{w} = 1.09$

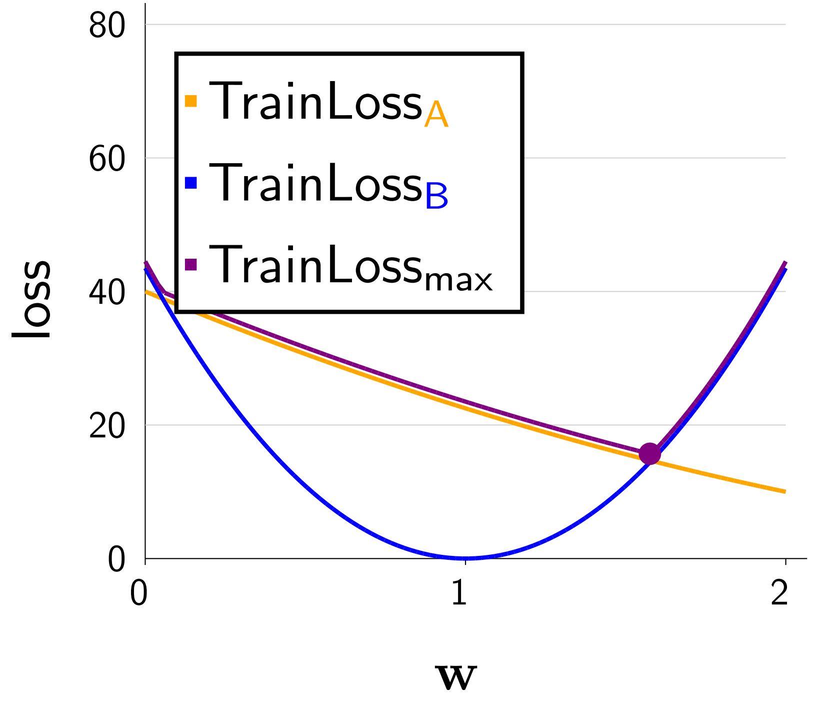
Group distributionally robust optimization (group DRO):

minimizer of maximum group loss: $\mathbf{w} = 1.58$

- Let us now compare the old average loss (the standard training loss) TrainLoss and the new maximum group loss $\text{TrainLoss}_{\text{max}}$.
- We minimize the average loss by setting the slope $\mathbf{w} = [1.09]$, yielding an average loss of 7.29. However, this setting gets much higher maximum group loss (over 20). Pictorially, we see that this $\mathbf{w}$ heavily favors the majority group (B).
- If instead we minimize the maximum group loss directly, we would choose $\mathbf{w} = 1.58$, yielding a maximum group loss of 15.59. Pictorially, we see that this solution pays more attention to (is closer to) the A points.
- Intuitively, the average loss favors majority groups over minority groups, but the maximum group loss gives a stronger voice to the minority groups, and as we see here, their influence is felt to a greater extent.

Training via gradient descent

x	y	g
1	4	A
2	8	A
5	5	B
6	6	B
7	7	B
8	8	B



$$\text{TrainLoss}_{\text{max}}(\mathbf{w}) = \max_g \text{TrainLoss}_g(\mathbf{w})$$

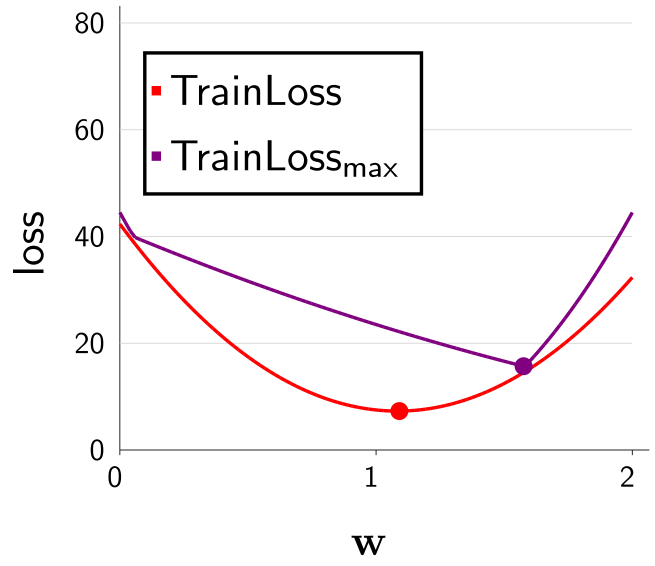
$$\nabla \text{TrainLoss}_{\text{max}}(\mathbf{w}) = \nabla \text{TrainLoss}_{g^*}(\mathbf{w})$$

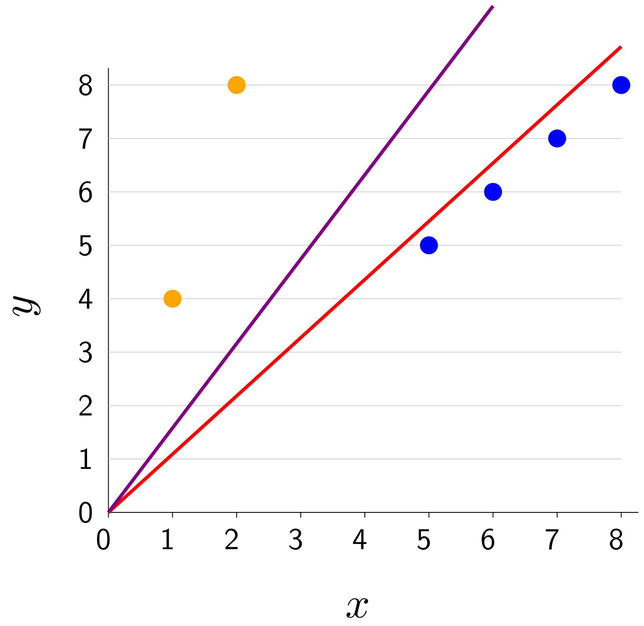
$$\text{where } g^* = \arg\max_g \text{TrainLoss}_g(\mathbf{w})$$

- In general, we can find minimize the maximum group loss by gradient descent.
- We just have to be able to take the gradient of $\text{TrainLoss}_{\text{max}}$, which is a maximum over the per-group losses.
- The gradient of a max is simply the gradient of the term that achieves the max.
- So algorithmically, it's very intuitive: you first compute whichever group (g^*) has the highest loss, and then you just evaluate the gradient only on per-group loss of that group (g^*).
- but you can see this paper for more details.

Summary

x	y	g
1	4	A
2	8	A
5	5	B
6	6	B
7	7	B
8	8	B





- Maximum group loss $\neq$ average loss
- Group DRO: minimize the maximum group loss
- Many more nuances: intersectionality? don't know groups? overfitting?

- To summarize, we've introduced the setting where examples are associated with groups. We see that by default, doing well on average is not the same as doing well on all groups (in other words, average loss is not the same as the maximum group loss), and the optimal solution for one objective is not optimal for the other objective.
- We presented an approach, group DRO that can ensure that the worst-off group is doing well.
- One parting remark is that while group DRO offers a mathematically clean solution, there are many subtleties when applying this to the real world. Intersectionality refers to the fact that groups which are defined by the conjunction of multiple attributes (e.g., White woman) may behave differently than the groups defined by individual attributes. What if you don't even have group information? How do you deal with overfitting (we've only looked at the training loss)?
- These are questions that are beyond the scope of this module, but I hope this short module has raised the need to think about about inequality as a first-class citizen, and piqued your interest to learn more.
- For further reading, consider checking out the book Fairness and machine learning: Limitations and Opportunities,

Roadmap

Group DRO

Non-linear features (offline)

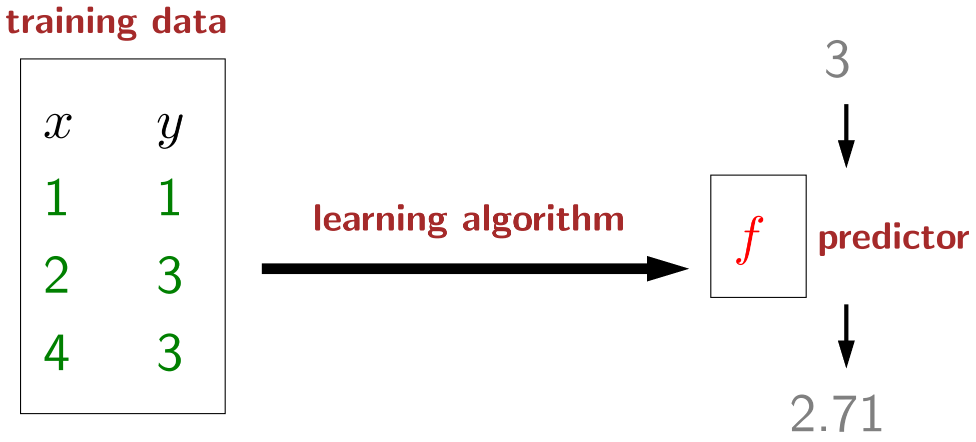
Feature templates (offline)

Neural networks

Backpropagation

- In this module, we'll show that even using the machinery of **linear** models, we can obtain much more powerful **non-linear** predictors.

Linear regression



Which predictors are possible?

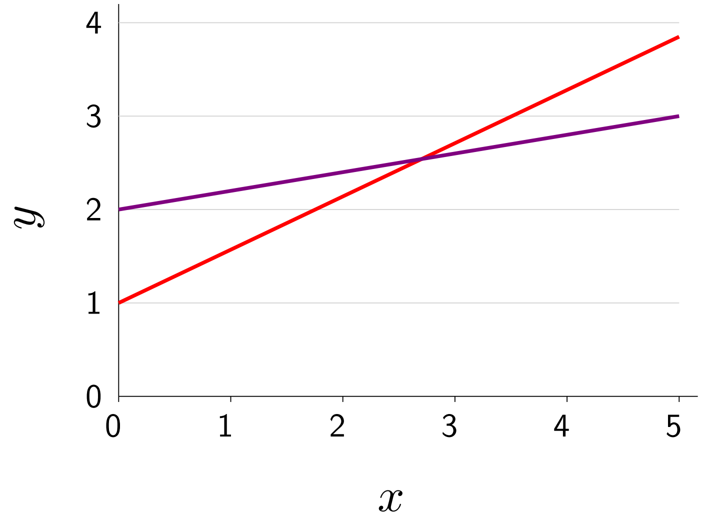
Hypothesis class

$$\mathcal{F} = \{f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x) : \mathbf{w} \in \mathbb{R}^d\}$$

$$\phi(x) = [1, x]$$

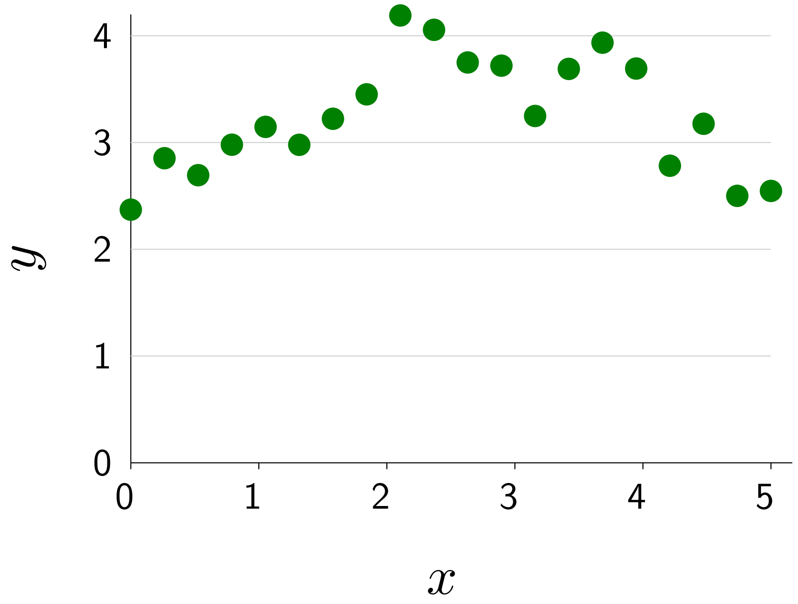
$$f(x) = [1, 0.57] \cdot \phi(x)$$

$$f(x) = [2, 0.2] \cdot \phi(x)$$



- We will look at regression and later turn to classification.
- Recall that in linear regression, given training data, a learning algorithm produces a predictor that maps new inputs to new outputs. The first design decision: what are the possible predictors that the learning algorithm can consider (what is the hypothesis class)?
- For linear predictors, remember the hypothesis class is the set of predictors that map some input x to the dot product between some weight vector $\mathbf{w}$ and the feature vector $\phi(x)$.
- As a simple example, if we define the feature extractor to be $\phi(x) = [1, x]$, then we can define various linear predictors with different intercepts and slopes.

More complex data



How do we fit a non-linear predictor?

- But sometimes data might be more complex and not be easily fit by a linear predictor. In this case, what can we do?
- One immediate reaction might be to go to something fancier like neural networks or decision trees.
- But let's see how far we can get with the machinery of linear predictors first.

Quadratic predictors

$$\phi(x) = [1, x, x^2]$$

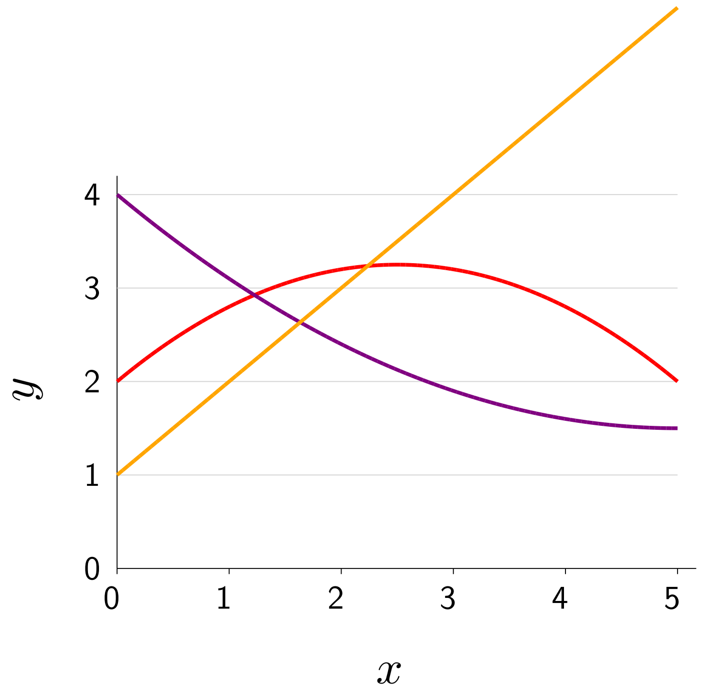
Example: $\phi(3) = [1, 3, 9]$

$$f(x) = [2, 1, -0.2] \cdot \phi(x)$$

$$f(x) = [4, -1, 0.1] \cdot \phi(x)$$

$$f(x) = [1, 1, 0] \cdot \phi(x)$$

$$\mathcal{F} = \{f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x) : \mathbf{w} \in \mathbb{R}^3\}$$



Non-linear predictors just by changing ϕ

- The key observation is that the feature extractor ϕ can be arbitrary.
- So let us define it to include an x^2 term.
- Now, by setting the weights appropriately, we can define a non-linear (specifically, a **quadratic**) predictor.
- The first two examples of quadratic predictors vary in intercept, slope and curvature.
- Note that by setting the weight for feature x^2 to zero, we recover linear predictors.
- Again, the hypothesis class is the set of all predictors $f_{\mathbf{w}}$ obtained by varying $\mathbf{w}$.
- Note that the hypothesis class of quadratic predictors is a **superset** of the hypothesis class of linear predictors.
- In summary, we've seen our first example of obtaining non-linear predictors just by changing the feature extractor ϕ!
- Advanced: here $x \in \mathbb{R}$ is one-dimensional, so x^2 is just one additional feature. If $x \in \mathbb{R}^d$ were d-dimensional, then there would be $O(d^2)$ quadratic features of the form $x_i x_j$ for $i, j \in \{1, \dots, d\}$. When d is large, then d^2 can be prohibitively large, which is one reason that using the machinery of linear predictors to increase expressivity can be problematic.

Piecewise constant predictors

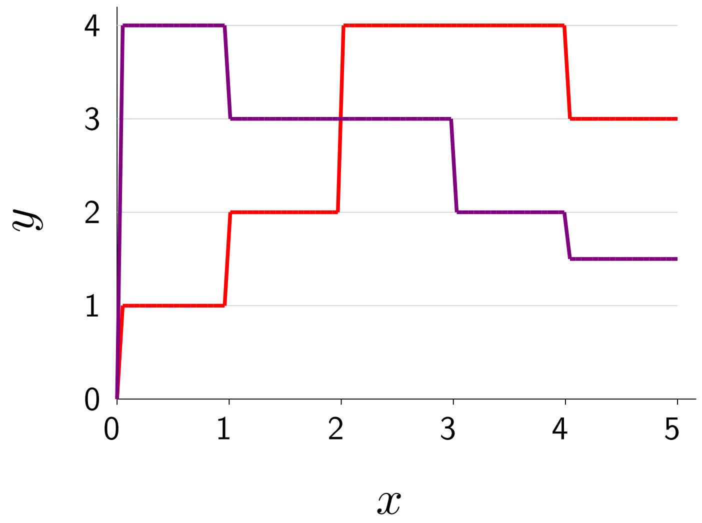
$$\phi(x) = [\mathbf{1}[0 < x \leq 1], \mathbf{1}[1 < x \leq 2], \mathbf{1}[2 < x \leq 3], \mathbf{1}[3 < x \leq 4], \mathbf{1}[4 < x \leq 5]]$$

Example: $\phi(2.3) = [0, 0, 1, 0, 0]$

$$f(x) = [1, 2, 4, 4, 3] \cdot \phi(x)$$

$$f(x) = [4, 3, 3, 2, 1.5] \cdot \phi(x)$$

$$\mathcal{F} = \{f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x) : \mathbf{w} \in \mathbb{R}^5\}$$



Expressive non-linear predictors by partitioning the input space

- Quadratic predictors are still a bit restricted: they can only go up and then down smoothly (or vice-vera).
- We introduce another type of feature extractor which divides the input space into regions and allows the predicted value of each region to vary independently, yielding piecewise constant predictors (see figure).
- Specifically, each component of the feature vector corresponds to one region (e.g., $[0, 1)$) and is 1 if x lies in that region and 0 otherwise.
- Assuming the regions are disjoint, the weight associated with a component/region is exactly the predicted value.
- As you make the regions smaller, then you have more features, and the expressivity of your hypothesis class increases. In the limit, you can essentially capture any predictor you want.
- Advanced: what happens if x were not a scalar, but a d-dimensional vector? Then if each component gets broken up into B bins, then there will be B^d features! For each feature, we need to fit its weight, and there will in generally be too few examples to fit all the features.

Predictors with periodicity structure

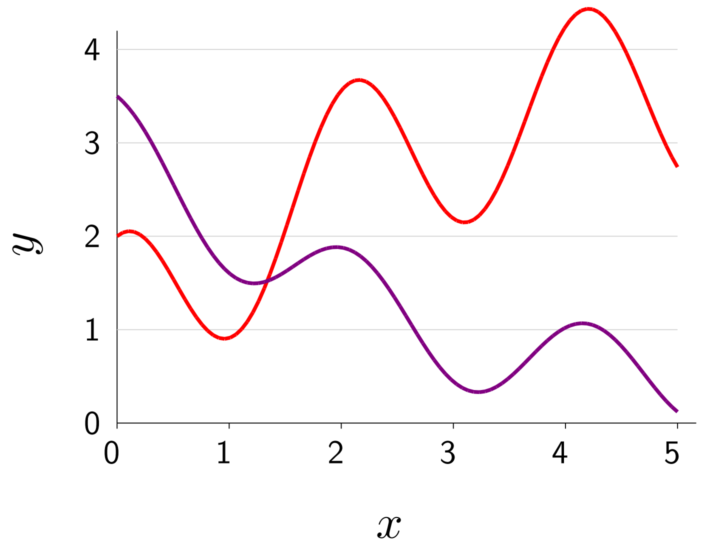
$$\phi(x) = [1, x, x^2, \cos(3x)]$$

Example: $\phi(2) = [1, 2, 4, 0.96]$

$$f(x) = [1, 1, -0.1, 1] \cdot \phi(x)$$

$$f(x) = [3, -1, 0.1, 0.5] \cdot \phi(x)$$

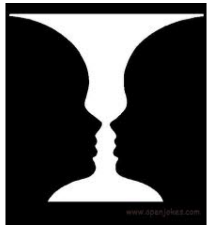
$$\mathcal{F} = \{f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x) : \mathbf{w} \in \mathbb{R}^4\}$$



Just throw in any features you want

- Quadratic and piecewise constant predictors are just two examples of an unboundedly large design space of possible feature extractors.
- Generally, the choice of features is informed by the prediction task that we wish to solve (either prior knowledge or preliminary data exploration).
- For example, if x represents time and we believe the true output y varies according to some periodic structure (e.g., traffic patterns repeat daily, sales patterns repeat annually), then we might use periodic features such as cosine to capture these trends.
- Each feature might represent some type of structure in the data. If we have multiple types of structures, these can just be "thrown in" into the feature vector.
- Features represent what properties **might** be useful for prediction. If a feature is not useful, then the learning algorithm can assign a weight close to zero to that feature. Of course, the more features one has, the harder learning becomes.

Linear in what?



Prediction:

$$f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x)$$

Linear in $\mathbf{w}$?	Yes
Linear in $\phi(x)$?	Yes
Linear in x?	No!

Key idea: non-linearity

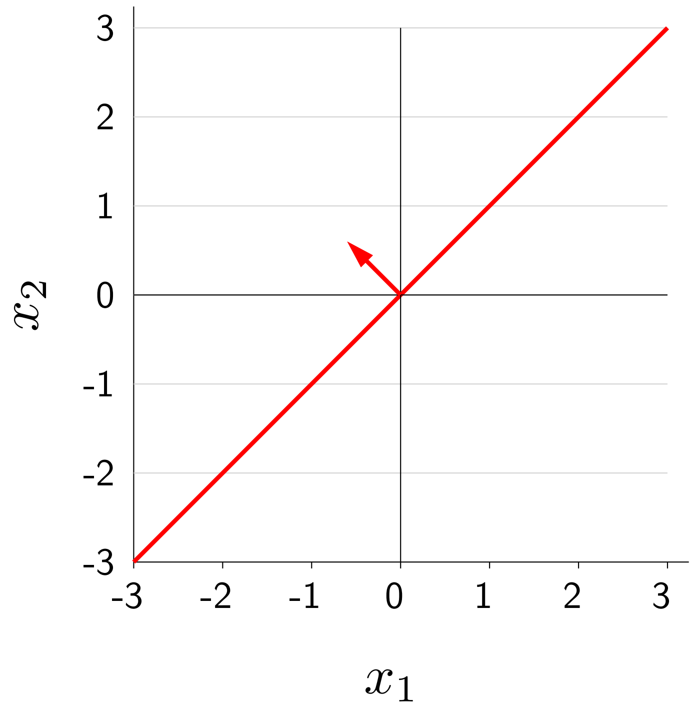
- Expressivity: score $\mathbf{w} \cdot \phi(x)$ can be a **non-linear** function of x
- Efficiency: score $\mathbf{w} \cdot \phi(x)$ always a **linear** function of $\mathbf{w}$

- Wait a minute...how are we able to obtain non-linear predictors if we're still using the machinery of linear predictors? It's a linguistic sleight of hand, as "linear" is ambiguous.
- The score is $\mathbf{w} \cdot \phi(x)$ linear in $\mathbf{w}$ and $\phi(x)$. However, the score is not linear in x (it might not even make sense because x need not be a vector at all — it could be a string or a PDF file).
- The significance is as follows: From the feature extractor's viewpoint, we can define arbitrary features that yield very **non-linear** functions in x.
- From the learning algorithm's viewpoint (which only looks at $\phi(x)$, not x), **linearity** us to optimize the weights efficiently.
- Advanced: if the score is linear in $\mathbf{w}$ and the loss function Loss is convex (which holds for the squared, hinge, logistic losses but not the zero-one loss), then minimizing the training loss TrainLoss is a convex optimization problem, and gradient descent with a proper step size is guaranteed to converge to the global minimum.

Linear classification

$$\phi(x) = [x_1, x_2]$$

$$f(x) = \text{sign}([-0.6, 0.6] \cdot \phi(x))$$



Decision boundary is a line

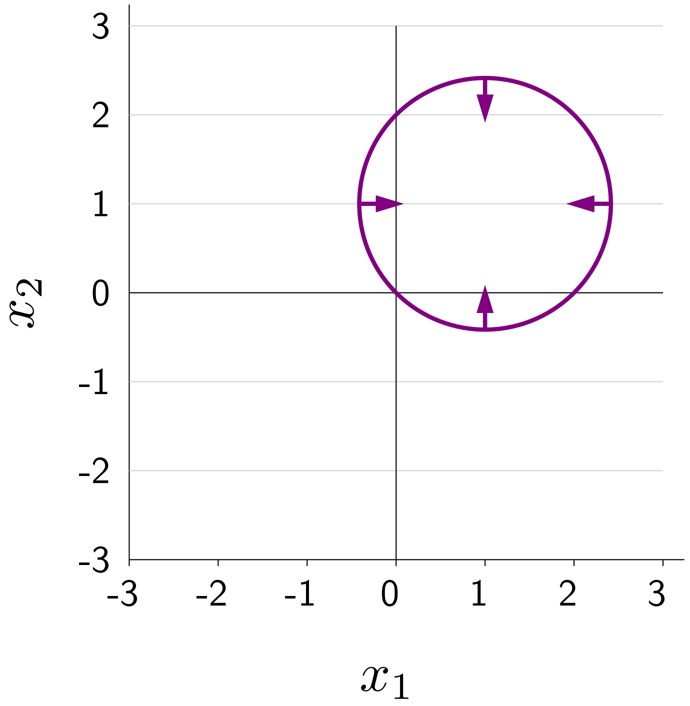
- Now let's turn from regression to classification.
- The story is pretty much the same: you can define arbitrary features to yield non-linear classifiers.
- Recall that in binary classification, the classifier (predictor) returns the sign of the score.
- The classifier can be therefore be represented by its decision boundary, which divides the input space into two regions: points with positive score and points with negative score.
- Note that the classifier $f_{\mathbf{w}}(x)$ is a non-linear function of x (and $\phi(x)$) no matter what (due to the sign function), so it is not helpful to talk about whether $f_{\mathbf{w}}$ is linear or non-linear. Instead we will ask whether the **decision boundary** corresponding to $f_{\mathbf{w}}$ is linear or not.

Quadratic classifiers

$$\phi(x) = [x_1, x_2, x_1^2 + x_2^2]$$
$$f(x) = \text{sign}([2, 2, -1] \cdot \phi(x))$$

Equivalently:

$$f(x) = \begin{cases} 1 & \text{if } \{(x\_1 - 1)^2 + (x\_2 - 1)^2 \le 2\} \\ -1 & \text{otherwise} \end{cases}$$



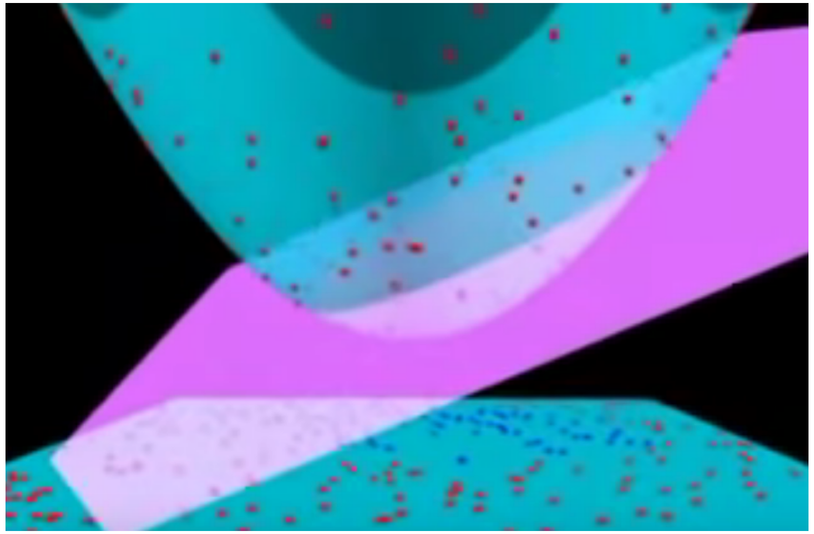
Decision boundary is a circle

- Let us see how we can define a classifier with a non-linear decision boundary.
- Let's try to construct a feature extractor that induces a decision boundary that is a circle: the inside is classified $+1$ and the outside is classified -1.
- We will add a new feature $x_1^2 + x_2^2$ into the feature vector, and define the weights to be as follows.
- Then rewrite the classifier to make it clear that it is the equation for the interior of a circle with radius $\sqrt{2}$.
- As a sanity check, we you can see that $x = [0, 0]$ results in a score of 0, which means that it is on the decision boundary. And as either of x_1 or x_2 grow in magnitude (either $|x_1| \to \infty$ or $|x_2| \to \infty$), the contribution of the third feature dominates and the sign of the score will be negative.

Visualization in feature space

Input space: $x = [x_1, x_2]$, decision boundary is a circle

Feature space: $\phi(x) = [x_1, x_2, x_1^2 + x_2^2]$, decision boundary is a hyperplane



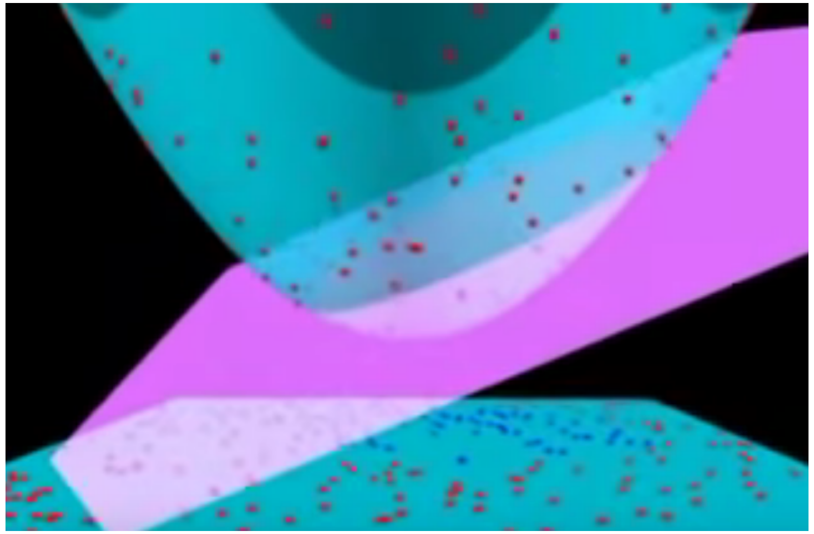
- Let's try to understand the relationship between the non-linearity in x and linearity in $\phi(x)$.
- Click on the image to see the linked video (which is about polynomial kernels and SVMs, but the same principle applies here).
- In the input space x, the decision boundary which separates the red and blue points is a circle.
- We can also visualize the points in **feature space**, where each point is given an additional dimension $x_1^2 + x_2^2$.
- In this three-dimensional feature space, a linear predictor (which is now defined by a hyperplane instead of a line) can in fact separate the red and blue points.
- This corresponds to the non-linear predictor in the original two-dimensional space.

Summary

$$f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x)$$

linear in $\mathbf{w}, \phi(x)$

non-linear in x



- Regression: non-linear predictor, classification: non-linear decision boundary
- Types of non-linear features: quadratic, piecewise constant, etc.

Non-linear predictors with linear machinery

- To summarize, we have shown that the term "linear" is ambiguous: a predictor in regression is non-linear in the input x but is linear in the feature vector $\phi(x)$.
- The score is also linear with respect to the weights $\mathbf{w}$, which is important for efficient learning.
- Classification is similar, except we talk about (non-)linearity of the decision boundary.
- We also saw many types of non-linear predictors that you could create by concocting various features (quadratic predictors, piecewise constant predictors).
- So next time someone on the street asks you about linear predictors, you should first ask them "linear in what?"

Roadmap

Group DRO

Non-linear features (offline)

Feature templates (offline)

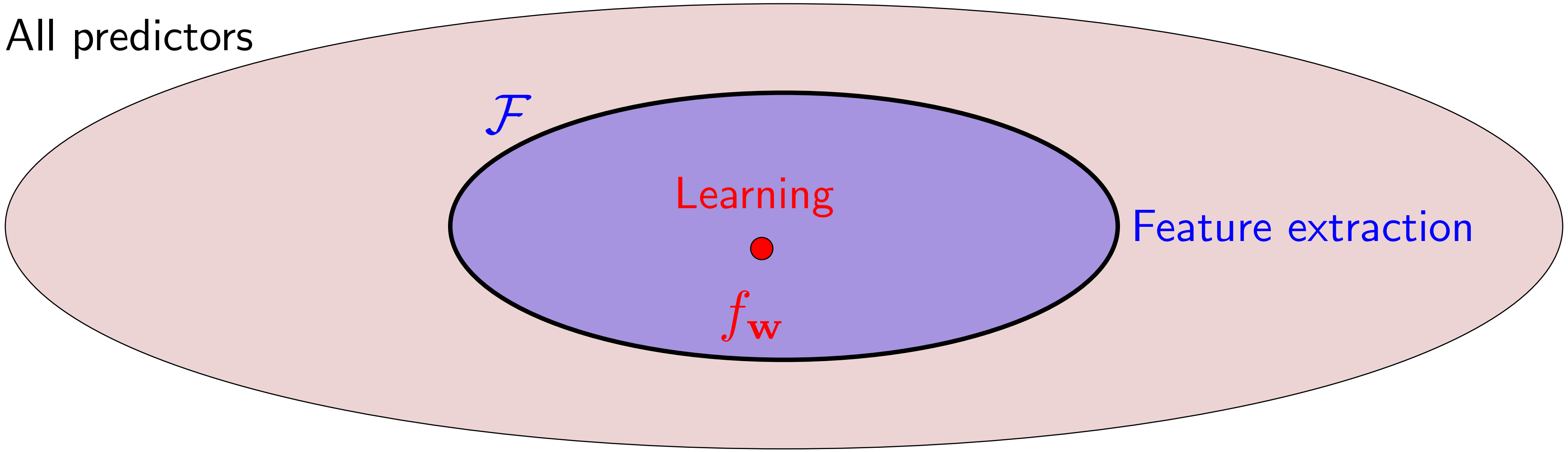
Neural networks

Backpropagation

- In this module, we'll talk about how to use feature templates to construct features in a flexible way.

Feature extraction + learning

$$\mathcal{F} = \{f_{\mathbf{w}}(x) = \text{sign}(\mathbf{w} \cdot \phi(x)) : \mathbf{w} \in \mathbb{R}^d\}$$



- Feature extraction: choose $\mathcal{F}$ based on domain knowledge
- Learning: choose $f_{\mathbf{w}} \in \mathcal{F}$ based on data

> Want $\mathcal{F}$ to contain good predictors but not be too big

- Recall that the hypothesis class $\mathcal{F}$ is the set of predictors considered by the learning algorithm. In the case of linear predictors, $\mathcal{F}$ is given by some function of $\mathbf{w} \cdot \phi(x)$ for all $\mathbf{w}$ (sign for classification, no sign for regression). This can be visualized as a set in the figure.
- Learning is the process of choosing a particular predictor $f_{\mathbf{w}}$ from $\mathcal{F}$ given training data.
- But the question that will concern us in this module is how do we choose $\mathcal{F}$? We saw some options already: linear predictors, quadratic predictors, etc., but what makes sense for a given application?
- If the hypothesis class doesn't contain any good predictors, then no amount of learning can help. So the question when extracting features is really whether they are powerful enough to **express** good predictors. It's okay and expected that $\mathcal{F}$ will contain bad ones as well. Of course, we don't want $\mathcal{F}$ to be too big, or else learning becomes hard, not just computationally but statistically (as we'll explain when we talk about generalization).

Feature extraction with feature names

Example task:

classifier

string (x) $\longrightarrow$ $\boxed{f_{\mathbf{w}}(x) = \text{sign}(\mathbf{w} \cdot \phi(x))}$ $\longrightarrow$ valid email address? (y)

Question: what properties of x **might be** relevant for predicting y?

Feature extractor: Given x, produce set of (feature name, feature value) pairs

abc@gmail.com

x

$$\xrightarrow[\text{arbitrary!}]{\textbf{feature extractor } \phi}$$

length>10	:	1
fracOfAlpha	:	0.85
contains_@	:	1
endsWith_com	:	1
endsWith_org	:	0

$\phi(x)$

- To get some intuition about feature extraction, let us consider the task of predicting whether whether a string is a valid email address or not.
- We will assume the classifier $f_{\mathbf{w}}$ is a linear classifier, which is given by some feature extractor ϕ.
- Feature extraction is a bit of an art that requires intuition about both the task and also what machine learning algorithms are capable of. The general principle is that features should represent properties of x which **might be** relevant for predicting y.
- Think about the feature extractor as producing a set of (feature name, feature value) pairs. For example, we might extract information about the length, or fraction of alphanumeric characters, whether it contains various substrings, etc.
- It is okay to add features which turn out to be irrelevant, since the learning algorithm can always in principle choose to ignore the feature, though it might take more data to do so.
- We have been associating each feature with a name so that it's easier for us (humans) to interpret and develop the feature extractor. The feature names act like the analogue of **comments** in code. Mathematically, the feature name is not needed by the learning algorithm and erasing them does not change prediction or learning.

Prediction with feature names

Weight vector $\mathbf{w} \in \mathbb{R}^d$

length>10	:	-1.2
fracOfAlpha	:	0.6
contains_@	:	3
endsWith_com	:	2.2
endsWith_org	:	1.4

Feature vector $\phi(x) \in \mathbb{R}^d$

length>10	:	1
fracOfAlpha	:	0.85
contains_@	:	1
endsWith_com	:	1
endsWith_org	:	0

Score: weighted combination of features

$$\mathbf{w} \cdot \phi(x) = \sum_{j=1}^{d} w_j \phi(x)_j$$

Example: $-1.2(1) + 0.6(0.85) + 3(1) + 2.2(1) + 1.4(0) = 4.51$

- A feature vector formally is just a list of numbers, but we have endowed each feature in the feature vector with a name.
- The weight vector is also just a list of numbers, but we can endow each weight with the corresponding name as well.
- Recall that the score is simply the dot product between the weight vector and the feature vector. In other words, the score aggregates the contribution of each feature, weighted appropriately.
- Each feature weight w_j determines how the corresponding feature value $\phi_j(x)$ contributes to the prediction.
- If w_j is positive, then the presence of feature j ($\phi_j(x) = 1$) favors a positive classification (e.g., ending with com). Conversely, if w_j is negative, then the presence of feature j favors a negative classification (e.g., length greater than 10). The magnitude of w_j measures the strength or importance of this contribution.
- Advanced: while tempting, it can be a bit misleading to interpret feature weights in isolation, because the learning algorithm treats $\mathbf{w}$ holistically. In particular, a feature weight w_j produced by a learning algorithm will change depending on the presence of other features. If the weight of a feature is positive, it doesn't necessarily mean that feature is positively correlated with the label.

Organization of features?

abc@gmail.com

x

feature extractor ϕ

arbitrary!

length>10	:	1
fracOfAlpha	:	0.85
contains_@	:	1
endsWith_com	:	1
endsWith_org	:	0

$\phi(x)$

Which features to include? Need an organizational principle...

- How would we go about about creating good features?
- Here, we used our prior knowledge to define certain features (contains_@) which we believe are helpful for detecting email addresses.
- But this is ad-hoc, and it's easy to miss useful features (e.g., endsWith_us), and there might be other features which are predictive but not intuitive.
- We need a more systematic way to go about this.

Feature templates

Definition: feature template

A **feature template** is a group of features all computed in a similar way.

abc@gmail.com → last three characters equals ___

endsWith_aaa	:	0
endsWith_aab	:	0
endsWith_aac	:	0
...		
endsWith_com	:	1
...		
endsWith_zzz	:	0

Define types of pattern to look for, not particular patterns

- A useful organization principle is a **feature template**, which groups all the features which are computed in a similar way. (People often use the word "feature" when they really mean "feature template".)
- Rather than defining individual features like endsWith_com, we can define a single feature template which expands into all the features that computes whether the input x matches any three characters.
- Typically, we will write a feature template as an English description with a blank (__), which is to be filled in with an arbitrary value.
- The upshot is that we don't need to know which particular patterns (e.g., three-character suffixes) are useful, but only that **existence** of certain patterns (e.g., three-character suffixes) are useful cue to look at.
- It is then up to the learning algorithm to figure out which patterns are useful by assigning the appropriate feature weights.

Feature templates example 1

Input:

abc@gmail.com

Feature template	Example feature	
Last three characters equals ___	Last three characters equals *com*	: 1
Length greater than ___	Length greater than *10*	: 1
Fraction of alphanumeric characters	Fraction of alphanumeric characters	: 0.85

- Here are some other examples of feature templates.
- Note that an isolated feature (e.g., fraction of alphanumeric characters) can be treated as a trivial feature template with no blanks to be filled.
- In many cases, the feature value is binary (0 or 1), but they can also be real numbers.

Feature templates example 2

Input:

Latitude: 37.4068176

Longitude: -122.1715122

Feature template	Example feature name	
Pixel intensity of image at row ___ and column ___ (___ channel)	Pixel intensity of image at row *10* and column *93* (*red* channel)	: 0.8
Latitude is in [___, ___] and longitude is in [___, ___]	Latitude is in [*37.4*, *37.5*] and longitude is in [*-122.2*, *-122.1*]	: 1

- As another example application, suppose the input is an aerial image along with the latitude/longitude corresponding where the image was taken. This type of input arises in poverty mapping and land cover classification.
- In this case, we might define one feature template corresponding to the pixel intensities at various pixel-wise row/column positions in the image across all the 3 color channels (e.g., red, green, blue).
- Another feature template might define a family of binary features, one for each region of the world, where each region is defined by a bounding box over latitude and longitude.

Sparsity in feature vectors

abc@gmail.com → last character equals ___

endsWith_a	:	0
endsWith_b	:	0
endsWith_c	:	0
endsWith_d	:	0
endsWith_e	:	0
endsWith_f	:	0
endsWith_g	:	0
endsWith_h	:	0
endsWith_i	:	0
endsWith_j	:	0
endsWith_k	:	0
endsWith_l	:	0
endsWith_m	:	**1**
endsWith_n	:	0
endsWith_o	:	0
endsWith_p	:	0
endsWith_q	:	0
endsWith_r	:	0
endsWith_s	:	0
endsWith_t	:	0
endsWith_u	:	0
endsWith_v	:	0
endsWith_w	:	0
endsWith_x	:	0
endsWith_y	:	0
endsWith_z	:	0

Compact representation:

```
{"endsWith_m": 1}
```

- In general, a feature template corresponds to many features, and sometimes, **for a given input**, most of the feature values are zero; that is, the feature vector is **sparse**.
- Of course, different feature vectors have different non-zero features.
- In this case, it would be inefficient to represent all the features explicitly. Instead, we can just store the values of the non-zero features, assuming all other feature values are zero by default.

Two feature vector implementations

Arrays (good for dense features):

pixelIntensity(0,0)	:	0.8
pixelIntensity(0,1)	:	0.6
pixelIntensity(0,2)	:	0.5
pixelIntensity(1,0)	:	0.5
pixelIntensity(1,1)	:	0.8
pixelIntensity(1,2)	:	0.7
pixelIntensity(2,0)	:	0.2
pixelIntensity(2,1)	:	0
pixelIntensity(2,2)	:	0.1

`[0.8, 0.6, 0.5, 0.5, 0.8, 0.7, 0.2, 0, 0.1]`

Dictionaries (good for sparse features):

fracOfAlpha	:	0.85
contains_a	:	0
contains_b	:	0
contains_c	:	0
contains_d	:	0
contains_e	:	0
...		
contains_@	:	1
...		

`{"fracOfAlpha": 0.85, "contains_@": 1}`

- In general, there are two common ways to implement feature vectors: using arrays and using dictionaries.
- **Arrays** assume a fixed ordering of the features and store the feature values as an array. This implementation is appropriate when the number of nonzeros is significant (the features are dense). Arrays are especially efficient in terms of space and speed (and you can take advantage of GPUs). In computer vision applications, features (e.g., the pixel intensity features) are generally dense, so arrays are more common.
- However, when we have sparsity (few nonzeros), it is typically more efficient to implement the feature vector as a **dictionary** (map) from strings to doubles rather than a fixed-size array of doubles. The features not in the dictionary implicitly have a default value of zero. This sparse implementation is useful for natural language processing with linear predictors, and is what allows us to work efficiently over millions of features. In Python, one would define a feature vector $\phi(x)$ as the dictionary `{"endsWith_"+x[-3:]: 1}`. Dictionaries do incur extra overhead compared to arrays, and therefore dictionaries are much slower when the features are not sparse.
- One advantage of the sparse feature implementation is that you don't have to instantiate all the set of possible features in advance; the weight vector can be initialized to $\{\}$, and only when a feature weight becomes non-zero do we store it. This means we can dynamically update a model with incrementally arriving data, which might instantiate new features.

Summary

$$\mathcal{F} = \{f_{\mathbf{w}}(x) = \text{sign}(\mathbf{w} \cdot \phi(x)) : \mathbf{w} \in \mathbb{R}^d\}$$

Feature template:

abc@gmail.com $\longrightarrow$ last three characters equals ___

endsWith_aaa	:	0
endsWith_aab	:	0
endsWith_aac	:	0
...		
endsWith_com	:	1
...		
endsWith_zzz	:	0

Dictionary implementation:

```
{"endsWith_com": 1}
```

- The question we are concerned with in this module is to how to define the hypothesis class $\mathcal{F}$, which in the case of linear predictors is the question of what the feature extractor ϕ is.
- We showed how **feature templates** can be useful for organizing the definition of many features, and that we can use dictionaries to represent **sparse** feature vectors efficiently.
- Stepping back, feature engineering is one of the most critical components in the practice of machine learning. It often does not get as much attention as it deserves, mostly because it is a bit of an art and somewhat domain-specific.
- More powerful predictors such as neural networks will alleviate some of the burden of feature engineering, but even neural networks use feature vectors as the initial starting point, and therefore its effectiveness is ultimately governed by how good the features are.

Roadmap

Group DRO

Non-linear features (offline)

Feature templates (offline)

Neural networks

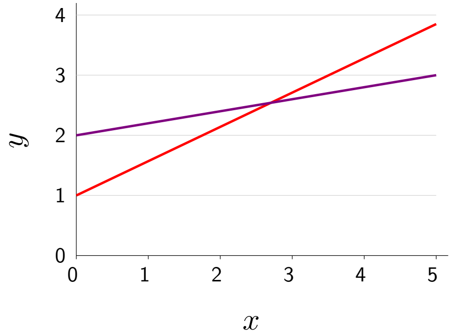
Backpropagation

- In this module, I will present neural networks, a way to construct non-linear predictors via problem decomposition.

Non-linear predictors

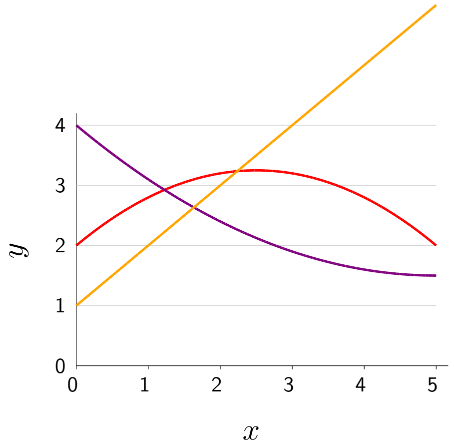
Linear predictors:

$$f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x), \quad \phi(x) = [1, x]$$



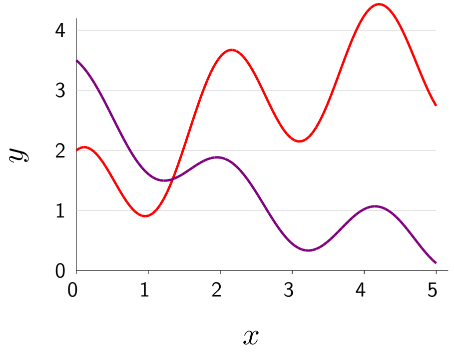
Non-linear (quadratic) predictors:

$$f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x), \quad \phi(x) = [1, x, x^2]$$



Non-linear neural networks:

$$f_{\mathbf{w}}(x) = \mathbf{w} \cdot \sigma(\mathbf{V}\phi(x)), \quad \phi(x) = [1, x]$$



- Recall that our first hypothesis class was linear (in x) predictors, which for regression means that the predictors are lines.
- However, we also showed that you could get non-linear (in x) predictors by simply changing the feature extractor ϕ. For example, by adding the feature x^2, one obtains quadratic predictors.
- One disadvantage of this approach is that if x were d-dimensional, one would need $O(d^2)$ features and corresponding weights, which presents considerable computational and statistical challenges.
- We will show that with neural networks, we can leave the feature extractor alone, but increase the complexity of predictor, which can also produce non-linear (though not necessarily quadratic) predictors.
- It is a common misconception that neural networks allow you to express more complex predictors. You can define ϕ to include essentially all predictors (as is done in kernel methods).
- Rather, neural networks yield non-linear predictors in a more **compact** way. For instance, you might not need $O(d^2)$ features to represent the desired non-linear predictor.

Motivating example

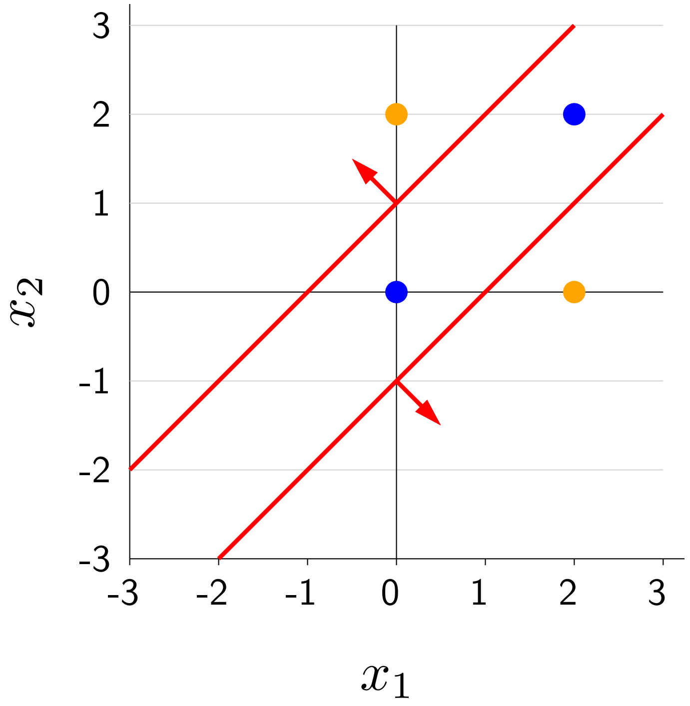
Example: predicting car collision

Input: positions of two oncoming cars $x = [x_1, x_2]$

Output: whether safe $(y = +1)$ or collide $(y = -1)$

Unknown: safe if cars sufficiently far: $y = \text{sign}(|x_1 - x_2| - 1)$

x_1	x_2	y
0	2	1
2	0	1
0	0	-1
2	2	-1



- As a motivating example, consider the problem of predicting whether two cars are going to collide given the their positions (as measured from distance from one side of the road). In particular, let x_1 be the position of one car and x_2 be the position of the other car.
- Suppose the true output is 1 (safe) whenever the cars are separated by a distance of at least 1. This relationship can be represented by the decision boundary which labels all points in the interior region between the two red lines as negative, and everything on the exterior (on either side) as positive. Of course, this true input-output relationship is unknown to the learning algorithm, which only sees training data. Consider a simple training dataset consisting of four points. (This is essentially the famous XOR problem that was impossible to fit using linear classifiers.)

Decomposing the problem

Test if car 1 is far right of car 2:

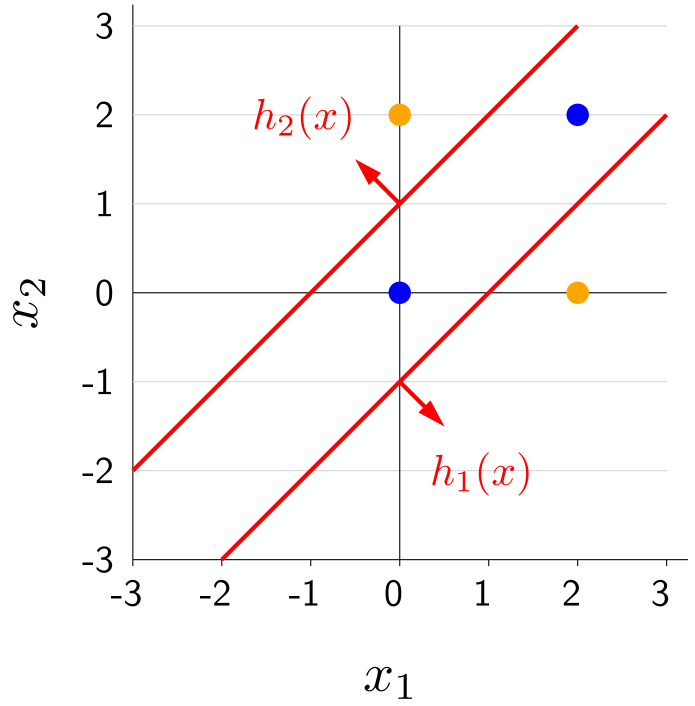
$$h_1(x) = \mathbf{1}[x_1 - x_2 \geq 1]$$

Test if car 2 is far right of car 1:

$$h_2(x) = \mathbf{1}[x_2 - x_1 \geq 1]$$

Safe if at least one is true:

$$f(x) = \text{sign}(h_1(x) + h_2(x))$$



x	$h_1(x)$	$h_2(x)$	$f(x)$
$[0,2]$	0	1	$+1$
$[2,0]$	1	0	$+1$
$[0,0]$	0	0	-1
$[2,2]$	0	0	-1

- One way to motivate neural networks (without appealing to the brain) is **problem decomposition**.
- The intuition is to break up the full problem into two subproblems: the first subproblem tests if car 1 is to the far right of car 2; the second subproblem tests if car 2 is to the far right of car 1. Then the final output is 1 iff at least one of the two subproblems returns 1.
- Concretely, we can define $h_1(x)$ to be the output of the first subproblem, which is a simple linear decision boundary (in fact, the right line in the figure).
- Analogously, we define $h_2(x)$ to be the output of the second subproblem.
- Note that $h_1(x)$ and $h_2(x)$ take on values 0 or 1 instead of -1 or $+1$.
- The points can then be classified by first computing $h_1(x)$ and $h_2(x)$, and then combining the results into $f(x)$.

Rewriting using vector notation

Intermediate subproblems:

$$h_1(x) = \mathbf{1}[x_1 - x_2 \geq 1] \;\; = \mathbf{1}[[-1, +1, -1] \cdot [1, x_1, x_2] \geq 0]$$

$$h_2(x) = \mathbf{1}[x_2 - x_1 \geq 1] \;\; = \mathbf{1}[[-1, -1, +1] \cdot [1, x_1, x_2] \geq 0]$$

$$\mathbf{h}(x) = \mathbf{1}\left[\begin{bmatrix} -1 & +1 & -1 \\ -1 & -1 & +1 \end{bmatrix}\begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix} \geq 0\right]$$

Predictor:

$$f(x) = \mathrm{sign}(h_1(x) + h_2(x)) \;\; = \mathrm{sign}([1, 1] \cdot \mathbf{h}(x))$$

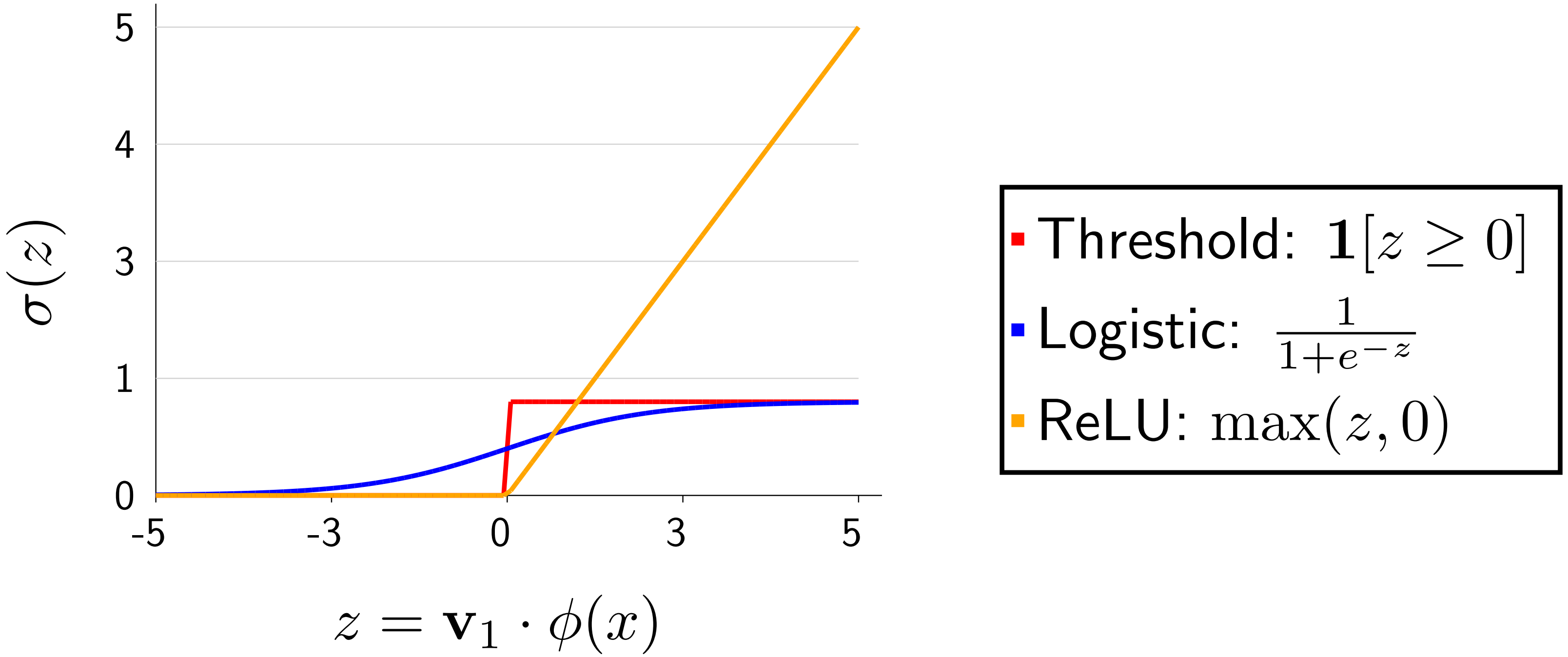
- Now let us rewrite this predictor $f(x)$ using vector notation.
- We can define a feature vector $[1, x_1, x_2]$ and a corresponding weight vector, where the dot product thresholded yields exactly $h_1(x)$.
- We do the same for $h_2(x)$.
- We put the two subproblems into one equation by stacking the weight vectors into one matrix. Recall that left-multiplication by a matrix is equivalent to taking the dot product with each row. By convention, the thresholding at 0 ($\mathbf{1}[\cdot \geq 0]$) applies component-wise.
- Finally, we can define the predictor in terms of a simple dot product.
- Now of course, we don't know the weight vectors, but we can learn them from the training data!

Avoid zero gradients

Problem: gradient of $h_1(x)$ with respect to $\mathbf{v}_1$ is 0

$$h_1(x) = \mathbf{1}[\mathbf{v}_1 \cdot \phi(x) \geq 0]$$

Solution: replace with an **activation function** σ with non-zero gradients

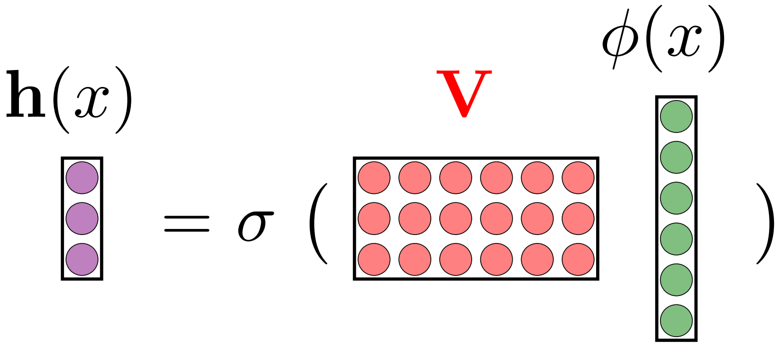


$$h_1(x) = \sigma(\mathbf{v}_1 \cdot \phi(x))$$

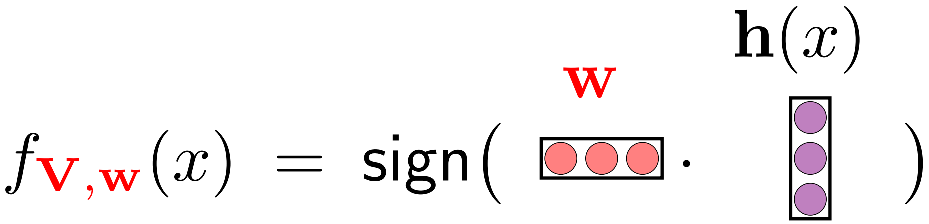
- Later we'll show how to perform learning using gradient descent, but we can anticipate one problem, which we encountered when we tried to optimize the zero-one loss.
- The gradient of $h_1(x)$ with respect to $\mathbf{v}_1$ is always zero because of the threshold function.
- To fix this, we replace the threshold function with an **activation function** with non-zero gradients
- Classically, neural networks used the **logistic function** $\sigma(z)$, which looks roughly like the threshold function but has non-zero gradients everywhere.
- Even though the gradients are non-zero, they can be quite small when $|z|$ is large (a phenomenon known as saturation). This makes optimizing with the logistic function still difficult.
- In 2012, Glorot et al. introduced the ReLU activation function, which is simply $\max(z, 0)$. This has the advantage that at least on the positive side, the gradient does not vanish (though on the negative side, the gradient is always zero). As a bonus, ReLU is easier to compute (only max, no exponentiation). In practice, ReLU works well and has become the activation function of choice.
- Note that if the activation function were linear (e.g., the identity function), then the gradients would always be nonzero, but you would lose the power of a neural network, because you would simply get the product of the final-layer weight vector and the weight matrix ($\mathbf{w}^\top \mathbf{V}$), which is equivalent to optimizing over a single weight vector.
- Therefore, that there is a tension between wanting an activation function that is non-linear but also has non-zero gradients.

Two-layer neural networks

Intermediate subproblems:



Predictor (classification):



Interpret $\mathbf{h}(x)$ as a learned feature representation!

Hypothesis class:

$$\mathcal{F} = \{f_{\mathbf{V},\mathbf{w}} : \mathbf{V} \in \mathbb{R}^{k \times d}, \mathbf{w} \in \mathbb{R}^{k}\}$$

- Now we are finally ready to define the hypothesis class of two-layer neural networks.
- We start with a feature vector $\phi(x)$.
- We multiply it by a weight matrix $\mathbf{V}$ (whose rows can be interpreted as the weight vectors of the k intermediate subproblems.
- Then we apply the activation function σ to each of the k components to get the hidden representation $\mathbf{h}(x) \in \mathbb{R}^k$.
- We can actually interpret $\mathbf{h}(x)$ as a learned feature vector (representation), which is derived from the original non-linear feature vector $\phi(x)$.
- Given $\mathbf{h}(x)$, we take the dot product with a weight vector $\mathbf{w}$ to get the score used to drive either regression or classification.
- The hypothesis class is the set of all such predictors obtained by varying the first-layer weight matrix $\mathbf{V}$ and the second-layer weight vector $\mathbf{w}$.

Deep neural networks

1-layer neural network:

$$\text{score} = \mathbf{w} \cdot \phi(x)$$

2-layer neural network:

$$\text{score} = \mathbf{w} \cdot \sigma\left(\mathbf{V} \; \phi(x) \right)$$

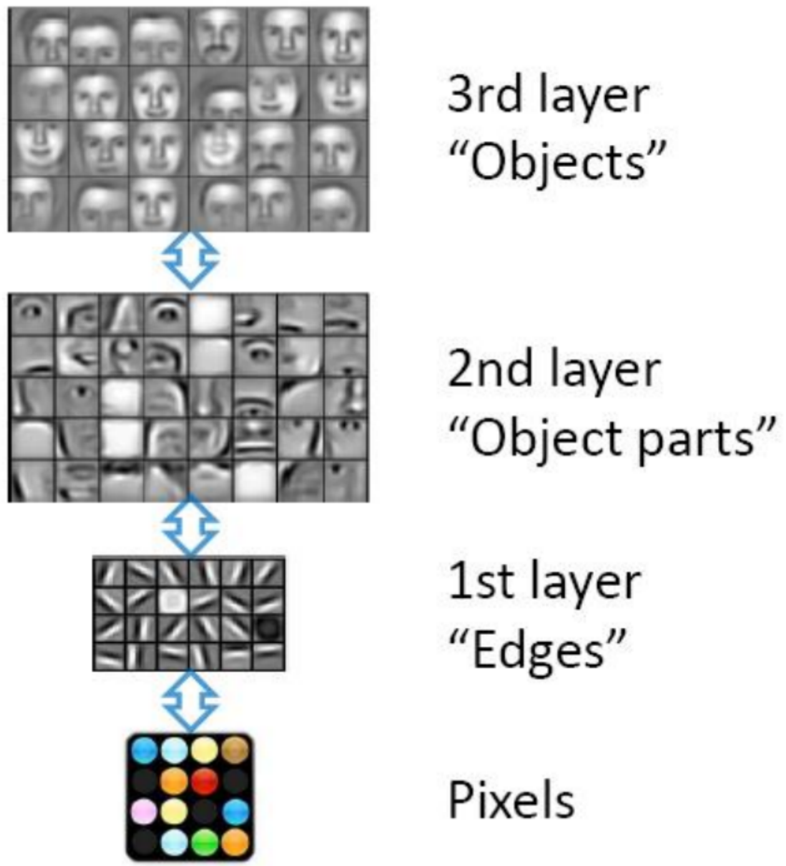
3-layer neural network:

$$\text{score} = \mathbf{w} \cdot \sigma\left(\mathbf{V}_2 \, \sigma\left(\mathbf{V}_1 \; \phi(x) \right) \right)$$

- We can push these ideas to build deep neural networks, which are neural networks with many layers.
- Warm up: for a one-layer neural network (a.k.a. a linear predictor), the score that drives prediction is simply a dot product between a weight vector and a feature vector.
- We just saw for a two-layer neural network, we apply a linear layer $\mathbf{V}$ first, followed by a non-linearity σ, and then take the dot product.
- To obtain a three-layer neural network, we apply a linear layer and a non-linearity (this is the basic building block). This can be iterated any number of times. No matter now deep the neural network is, the top layer is always a linear function, and all the layers below that can be interpreted as defining a (possibly very complex) hidden feature vector.
- In practice, you would also have a bias term (e.g., $\mathbf{V}\phi(x) + b$). We have omitted all bias terms for notational simplicity.

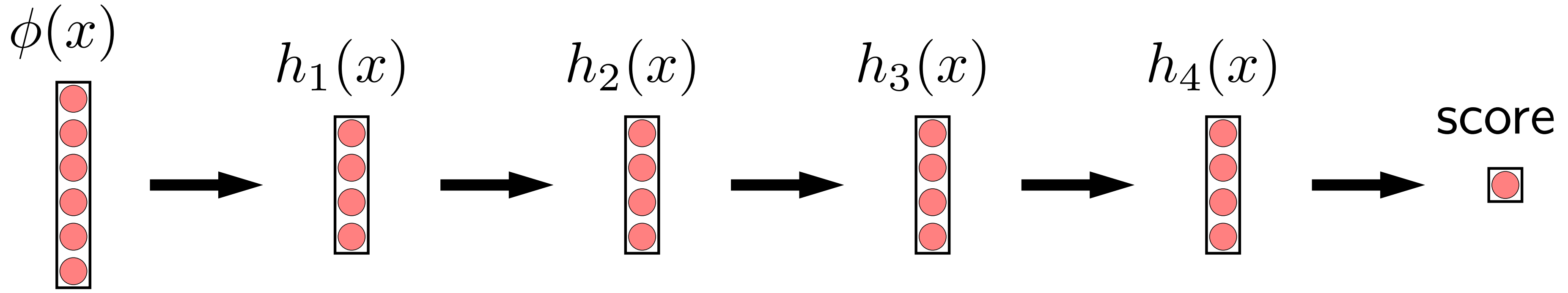
[figure from Honglak Lee]

Layers represent multiple levels of abstractions



- It can be difficult to understand what a sequence of (matrix multiply, non-linearity) operations buys you.
- To provide intuition, suppose the input feature vector $\phi(x)$ is a vector of all the pixels in an image.
- Then each layer can be thought of producing an increasingly abstract representation of the input. The first layer detects edges, the second detects object parts, the third detects objects. What is shown in the figure is for each component j of the hidden representation $\mathbf{h}(x)$, the input image $\phi(x)$ that maximizes the value of $h_j(x)$.
- Though we haven't talked about learning neural networks, it turns out that the "levels of abstraction" story is actually borne out visually when we learn neural networks on real data (e.g., images).

Why depth?

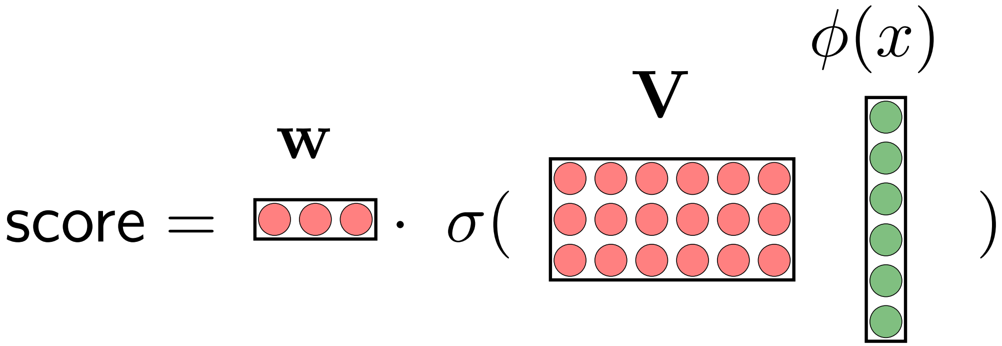


Intuitions:

- Multiple levels of abstraction
- Multiple steps of computation
- Empirically works well
- Theory is still incomplete

- Beyond learning hierarchical feature representations, deep neural networks can be interpreted in a few other ways.
- One perspective is that each layer can be thought of as performing some computation, and therefore deep neural networks can be thought of as performing multiple steps of computation.
- But ultimately, the real reason why deep neural networks are interesting is because they work well in practice.
- From a theoretical perspective, we have a quite an incomplete explanation for why depth is important. The original motivation from McCulloch/Pitts in 1943 showed that neural networks can be used to simulate a bounded computation logic circuit. Separately it has been shown that depth $k+1$ logic circuits can represent more functions than depth k. However, neural networks are real-valued and might have types of computations which don't fit neatly into logical paradigm. Obtaining a better theoretical understanding is an active area of research in statistical learning theory.

Summary



- Intuition: decompose problem into intermediate parallel subproblems
- Deep networks iterate this decomposition multiple times
- Hypothesis class contains predictors ranging over weights for all layers
- Next up: learning neural networks

- To summarize, we started with a toy problem (the XOR problem) and used it to motivate neural networks, which decompose a problem into intermediate subproblems, which are solved in parallel.
- Deep networks iterate this multiple times to build increasingly high-level representations of the input.
- Next, we will see how we can learn a neural network by choosing the weights for all the layers.

Roadmap

Group DRO

Non-linear features (offline)

Feature templates (offline)

Neural networks

Backpropagation

- In this module, I'll discuss **backpropagation**, an algorithm to automatically compute gradients.
- It is generally associated with training neural networks, but actually it is much more general and applies to any function.

Motivation: regression with four-layer neural networks

Loss on one example:

$$\text{Loss}(x, y, \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{w}) = (\mathbf{w} \cdot \sigma(\mathbf{V}_3 \sigma(\mathbf{V}_2 \sigma(\mathbf{V}_1 \phi(x)))) - y)^2$$

Stochastic gradient descent:

$$\mathbf{V}_1 \leftarrow \mathbf{V}_1 - \eta \nabla_{\mathbf{V}_1} \text{Loss}(x, y, \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{w})$$

$$\mathbf{V}_2 \leftarrow \mathbf{V}_2 - \eta \nabla_{\mathbf{V}_2} \text{Loss}(x, y, \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{w})$$

$$\mathbf{V}_3 \leftarrow \mathbf{V}_3 - \eta \nabla_{\mathbf{V}_3} \text{Loss}(x, y, \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{w})$$

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla_{\mathbf{w}} \text{Loss}(x, y, \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{w})$$

How to get the gradient without doing manual work?

- So far, we've defined neural networks, which take an initial feature vector $\phi(x)$ and sends it through a sequence of matrix multiplications and non-linear activations σ. At the end, we take the dot product between a weight vector $\mathbf{w}$ to produce the score.
- In regression, we predict the score, and use the squared loss, which looks at the squared difference betwen the score and the target y.
- Recall that we can use stochastic gradient descent to optimize the training loss (which is an average over the per-example losses). Now, we need to update all the weight matrices, not just a single weight vector. This can be done by taking the gradient with respect to each weight vector/matrix separately, and updating the respective weight vector/matrix by subtracting the gradient times a step size η.
- We can now proceed to take the gradient of the loss function with respect to the various weight vector/matrices. You should know how to do this: just apply the chain rule. But grinding through this complex expression by hand can be quite tedious. If only we had a way for this to be done automatically for us...

Computation graphs

$$\text{Loss}(x, y, \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{w}) = (\mathbf{w} \cdot \sigma(\mathbf{V}_3 \sigma(\mathbf{V}_2 \sigma(\mathbf{V}_1 \phi(x)))) - y)^2$$

Definition: computation graph

A directed acyclic graph whose root node represents the final mathematical expression and each node represents intermediate subexpressions.

Upshot: compute gradients via general **backpropagation** algorithm

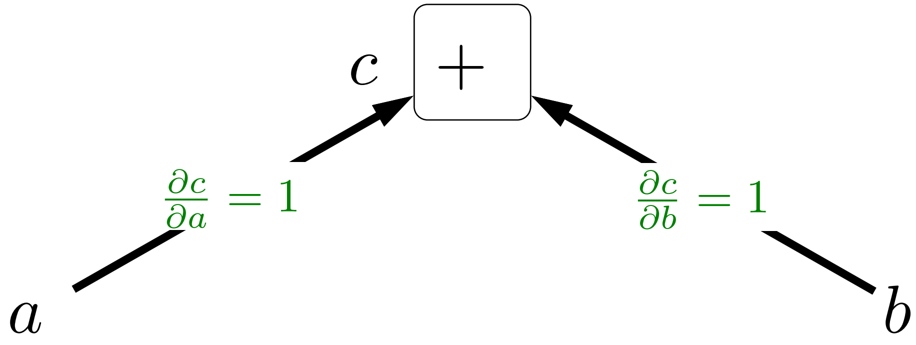
Purposes:

- Automatically compute gradients (how TensorFlow and PyTorch work)
- Gain insight into modular structure of gradient computations

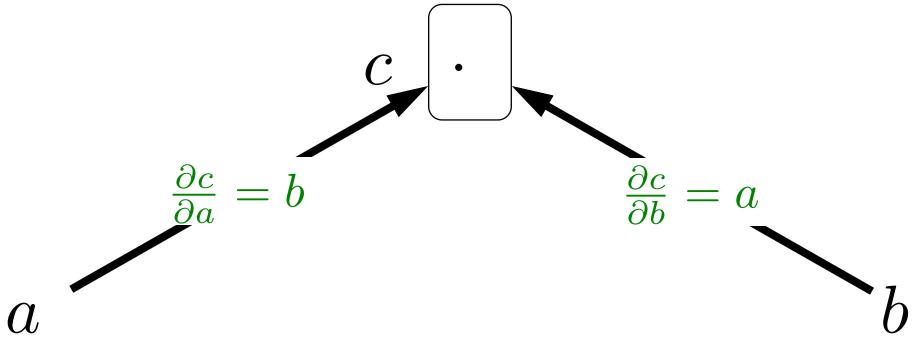
- Enter computation graphs, which will rescue us.
- A computation graph is a directed acyclic graph that represents an arbitrary mathematical expression. The root of that node represents the final expression, and the other nodes represent intermediate subexpressions.
- After having constructed the graph, we can compute all the gradients we want by running the general-purpose backpropagation algorithm, which operates on an arbitrary computation graph.
- There are two purposes to using computation graphs. The first and most obvious one is that it avoids having us to do pages of calculus, and instead delegates this to a computer. This is what packages such as TensorFlow or PyTorch do, and essentially all non-trivial deep learning models are trained like this.
- The second purpose is that by defining the graph, we can gain more insight into the nature of how gradients are computed in a modular way.

Functions as boxes

$$c = a + b$$



$$c = a \cdot b$$



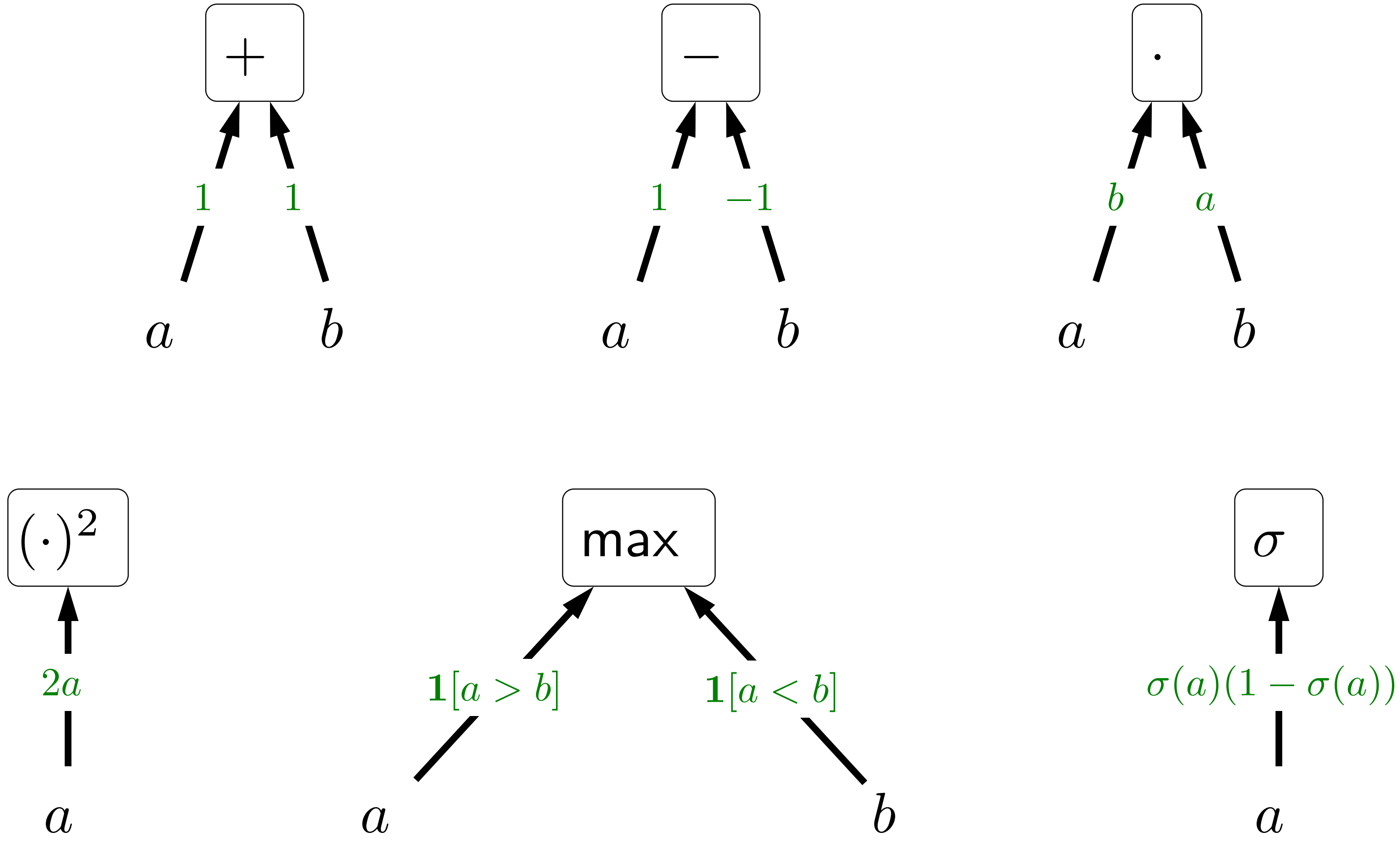
$$(a + \epsilon) + b = c + 1\epsilon$$
$$a + (b + \epsilon) = c + 1\epsilon$$

$$(a + \epsilon)b = c + b\epsilon$$
$$a(b + \epsilon) = c + a\epsilon$$

Gradients: how much does c change if a or b changes?

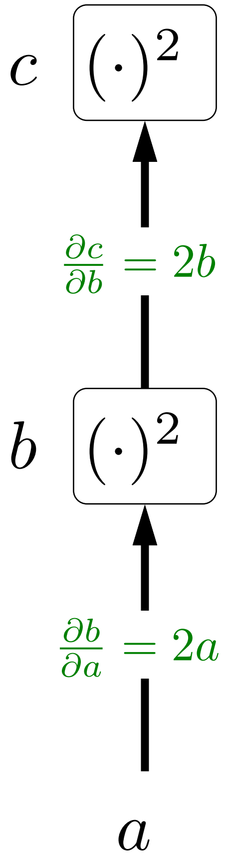
- The first conceptual step is to think of functions as boxes that take a set of inputs and produces an output.
- For example, take $c = a + b$. The key question is: if we perturb a by a small amount ϵ, how much does the output c change? In this case, the output c is also perturbed by 1ϵ, so the gradient (partial derivative) is 1. We put this gradient on the edge.
- We can handle $c = a \cdot b$ in a similar way.
- Intuitively, the gradient is a measure of local sensivity: how much input perturbations get amplified when they go through the various functions.

Basic building blocks



- Here are some more examples of simple functions and their gradients. Let's walk through them together.
- These should be familiar from basic calculus. All we've done is present them in a visually more intuitive way.
- For the max function, changing a only impacts the max iff $a > b$; and analogously for b.
- For the logistic function $\sigma(z) = \frac{1}{1+e^{-z}}$, a bit of algebraic elbow grease produces the gradient. You can check that the gradient is zero when $|a| \to \infty$.
- It turns out that these simple functions are all we need to build up many of the more complex and potentially scarier looking functions that we'll encounter.

Function composition

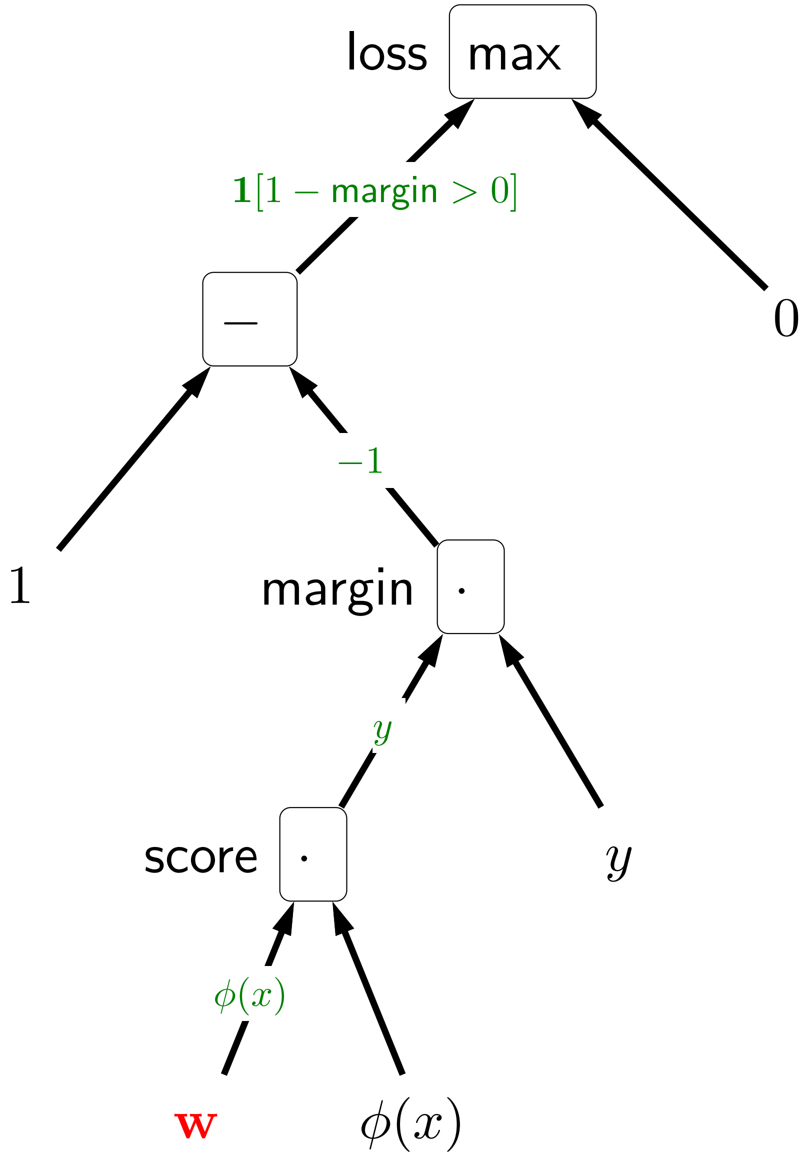


Chain rule:

$$\frac{\partial c}{\partial a} = \frac{\partial c}{\partial b}\frac{\partial b}{\partial a} = (2b)(2a) = (2a^2)(2a) = 4a^3$$

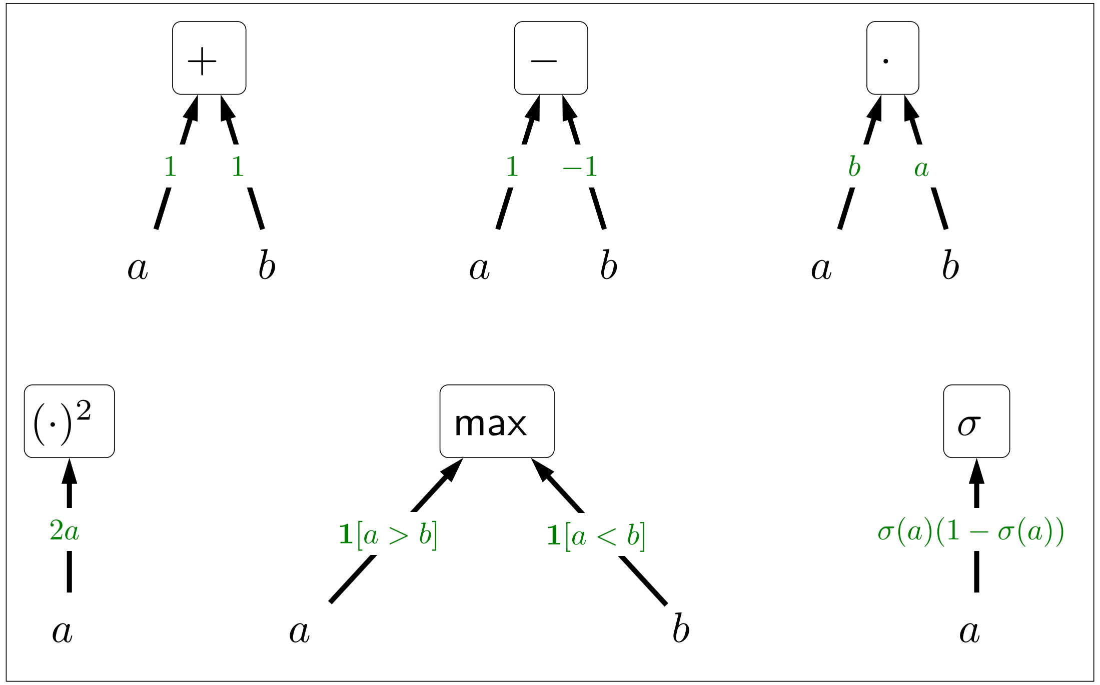
- Given these building blocks, we can now put them together to create more complex functions.
- Consider applying some function (e.g., squared) to a to get b, and then applying some other function (e.g., squared) to get c.
- What is the gradient of c with respect to a?
- We know from our building blocks the gradients on the edges.
- The final answer is given by the **chain rule** from calculus: just multiply the two gradients together.
- You can verify that this yields the correct answer $(2b)(2a) = 4a^3$.
- This visual intuition will help us better understand more complex functions.

Linear classification with hinge loss



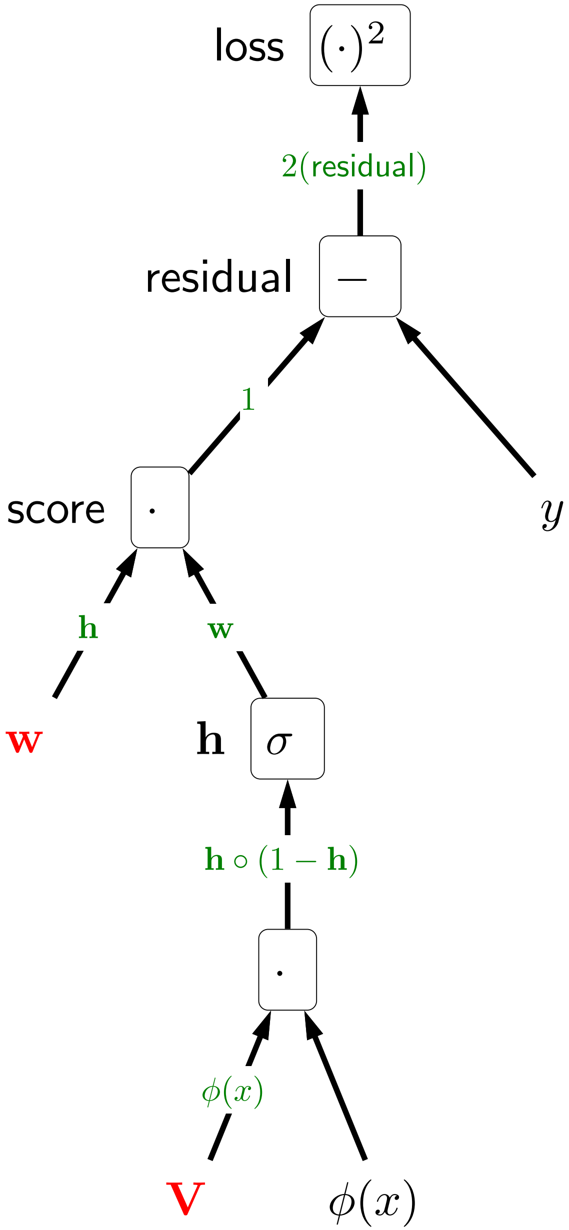
$$\text{Loss}(x, y, \mathbf{w}) = \max\{1 - \mathbf{w} \cdot \phi(x) y, 0\}$$

$$\nabla_{\mathbf{w}} \text{Loss}(x, y, \mathbf{w}) = -\mathbf{1}[\text{margin} < 1]\phi(x)y$$



- Now let's turn to our first real-world example: the hinge loss for linear classification. We already computed the gradient before, but let's do it using computation graphs.
- We can construct the computation graph for this expression, proceeding bottom up. At the leaves are the inputs and the constants. Each internal node is labeled with the operation (e.g., $\cdot$) and is labeled with a variable naming that subexpression (e.g., margin).
- In red, we have highlighted the weights $\mathbf{w}$ with respect to which we want to take the gradient. The central question is how small perturbations in $\mathbf{w}$ affect a change in the output (loss).
- We can examine each edge from the path from $\mathbf{w}$ to loss, and compute the gradient using our handy reference of building blocks.
- The actual gradient is the product of the edge-wise gradients from $\mathbf{w}$ to the loss output.

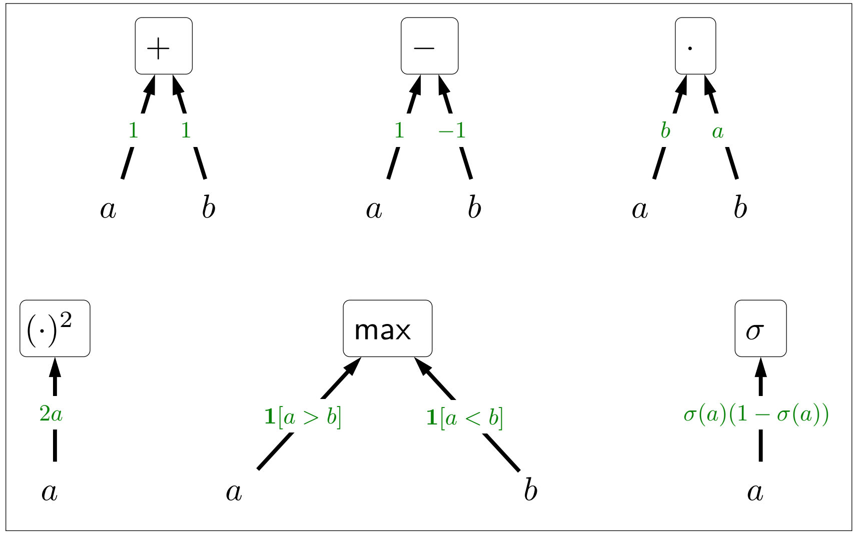
Two-layer neural networks



$$\text{Loss}(x, y, \mathbf{V}, \mathbf{w}) = (\mathbf{w} \cdot \sigma(\mathbf{V}\phi(x)) - y)^2$$

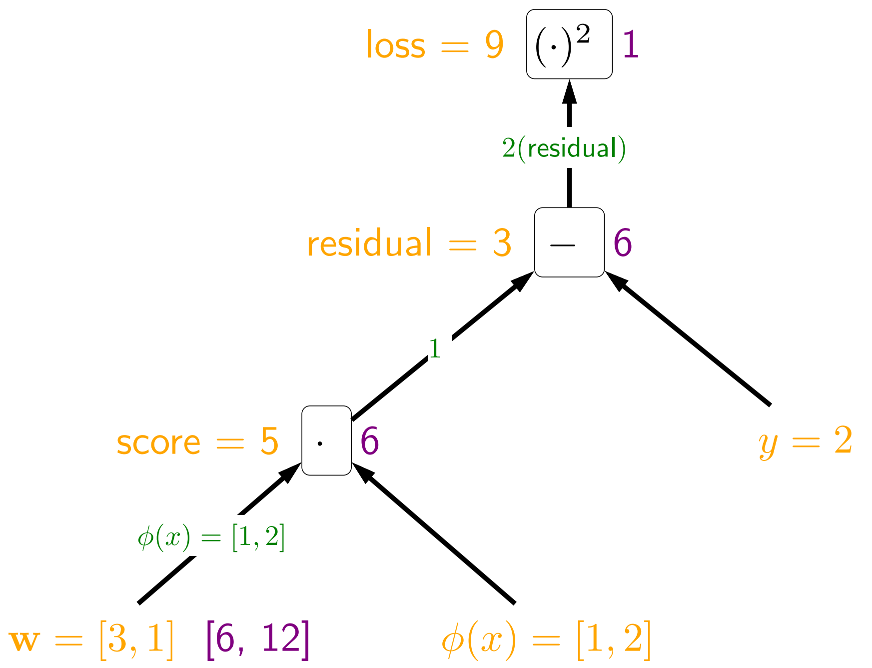
$$\nabla_{\mathbf{w}} \text{Loss}(x, y, \mathbf{V}, \mathbf{w}) = 2(\text{residual})\mathbf{h}$$

$$\nabla_{\mathbf{V}} \text{Loss}(x, y, \mathbf{V}, \mathbf{w}) = 2(\text{residual})\mathbf{w} \circ \mathbf{h} \circ (1 - \mathbf{h})\phi(x)^\top$$



- We now finally turn to neural networks, but the idea is essentially the same.
- Specifically, consider a two-layer neural network driving the squared loss.
- Let us build the computation graph bottom up.
- Now we need to take the gradient with respect to $\mathbf{w}$ and $\mathbf{V}$. Again, these are just the product of the gradients on the paths from $\mathbf{w}$ or $\mathbf{V}$ to the loss node at the root.
- Note that the two gradients have in common the the first two terms. Common paths result in common subexpressions for the gradient.
- There are some technicalities when dealing with vectors worth mentioning: First, the $\circ$ in $\mathbf{h} \circ (1 - \mathbf{h})$ is elementwise multiplication (not the dot product), since the non-linearity σ is applied elementwise. Second, there is a transpose for the gradient expression with respect to $\mathbf{V}$ and not $\mathbf{w}$ because we are taking $\mathbf{V}\phi(x)$, while taking $\mathbf{w} \cdot \mathbf{h} = \mathbf{w}^\top \mathbf{h}$.
- This computation graph also highlights the modularity of hypothesis class and loss function. You can pick any hypothesis class (linear predictors or neural networks) to drive the score, and the score can be fed into any loss function (squared, hinge, etc.).

Backpropagation



$$\text{Loss}(x, y, \mathbf{w}) = (\mathbf{w} \cdot \phi(x) - y)^2$$

$$\mathbf{w} = [3, 1], \phi(x) = [1, 2], y = 2$$

backpropagation

$$\nabla_{\mathbf{w}} \text{Loss}(x, y, \mathbf{w}) = [6, 12]$$

Definition: Forward/backward values

Forward: f_i is value for subexpression rooted at i

Backward: $g_i = \frac{\partial \text{loss}}{\partial f_i}$ is how f_i influences loss

Algorithm: backpropagation algorithm

Forward pass: compute each f_i (from leaves to root)

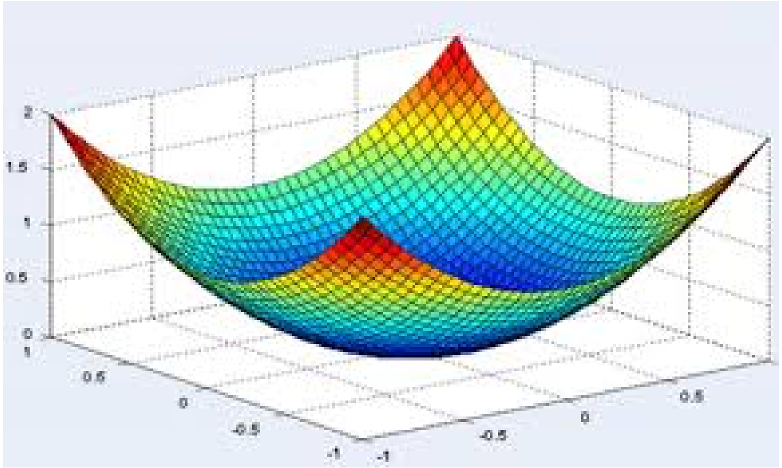
Backward pass: compute each g_i (from root to leaves)

- So far, we have mainly used the graphical representation to visualize the computation of function values and gradients for our conceptual understanding.
- Now let us introduce the **backpropagation** algorithm, a general procedure for computing gradients given only the specification of the function.
- Let us go back to the simplest example: linear regression with the squared loss.
- All the quantities that we've been computing have been so far symbolic, but the actual algorithm works on real numbers and vectors. So let's use concrete values to illustrate the backpropagation algorithm.
- The backpropagation algorithm has two phases: forward and backward. In the forward phase, we compute a **forward value** f_i for each node, coresponding to the evaluation of that subexpression. Let's work through the example.
- In the backward phase, we compute a **backward value** g_i for each node. This value is the gradient of the loss with respect to that node, which is also the product of all the gradients on the edges from the node to the root. To compute this backward value, we simply take the parent's backward value and multiply by the gradient on the edge to the parent. Let's work through the example.
- Note that both f_i and g_i can either be scalars, vectors, or matrices, but have the same dimensionality.

A note on optimization

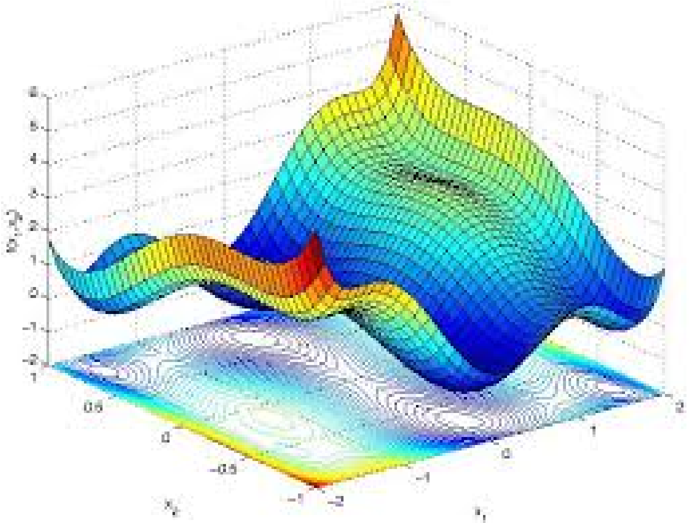
$$\min_{\mathbf{V},\mathbf{w}} \text{TrainLoss}(\mathbf{V}, \mathbf{w})$$

Linear predictors



(convex)

Neural networks



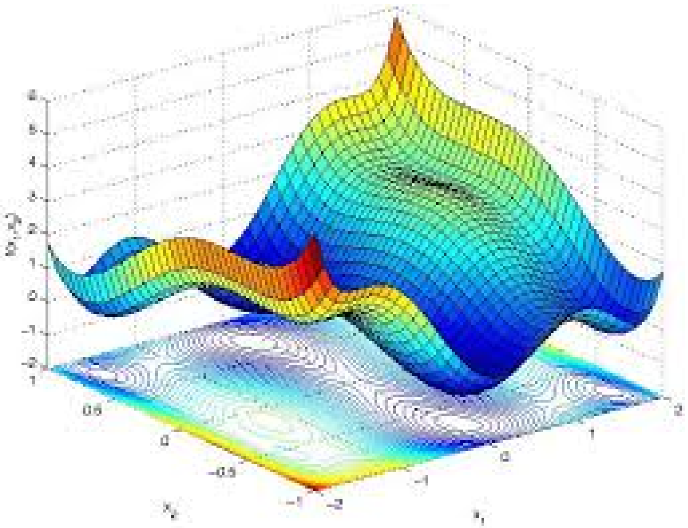
(non-convex)

Optimization of neural networks is in principle hard

- So now we can apply the backpropagation algorithm and compute gradients, stick them into stochastic gradient descent, and get some answer out.
- One question which we haven't addressed is whether stochastic gradient descent will work in the sense of actually finding the weights that minimize the training loss?
- For linear predictors (using the squared loss or hinge loss), $\text{TrainLoss}(\mathbf{w})$ is a convex function, which means that SGD (with an appropriately step size) is theoretically guaranteed to converge to the global optimum.
- However, for neural networks, $\text{TrainLoss}(\mathbf{V}, \mathbf{w})$ is typically non-convex which means that there are multiple local optima, and SGD is not guaranteed to converge to the global optimum. There are many settings that SGD fails both theoretically and empirically, but in practice, SGD on neural networks can work much better than theory would predict, provided certain precautions are taken. The gap between theory and practice is not well understood and an active area of research.

How to train neural networks

$$\text{score} = \mathbf{w} \cdot \sigma(\mathbf{V} \phi(x))$$

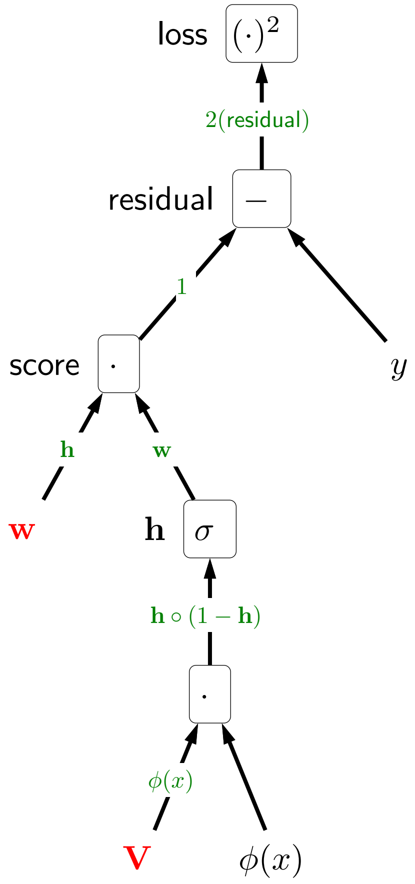


- Careful initialization (random noise, pre-training)
- Overparameterization (more hidden units than needed)
- Adaptive step sizes (AdaGrad, Adam)

Don't let gradients vanish or explode!

- Training a neural network is very much like driving stick. In practice, there are some "tricks" that are needed to make things work properly. Just to name a few to give you a sense of the considerations:
- Initialization (where you start the weights) matters for non-convex optimization. Unlike for linear models, you can't start at zero or else all the subproblems will be the same (all rows of $\mathbf{V}$ will be the same). Instead, you want to initialize with a small amount of random noise.
- It is common to use overparameterized neural networks, ones with more hidden units (k) than is needed, because then there are more "chances" that some of them will pick out on the right signal, and it is okay if some of the hidden units become "dead".
- There are small but important extensions of stochastic gradient descent that allow the step size to be tuned per weight.
- Perhaps one high-level piece of advice is that when training a neural network, it is important to monitor the gradients. If they vanish (get too small), then training won't make progress. If they explode (get too big), then training will be unstable.

Summary



- Computation graphs: visualize and understand gradients
- Backpropagation: general-purpose algorithm for computing gradients

- The most important concept in this module is the idea of a **computation graph**, allows us to represent arbitrary mathematical expressions, which can just be built out of simple building blocks. They hopefully have given you a more visual and better understanding of what gradients are about.
- The **backpropagation** algorithm allows us to simply write down an expression, and never have to take a gradient manually again. However, it is still important to understand how the gradient arises, so that when you try to train a deep neural network and your gradients vanish, you know how to think about debugging your network.
- The generality of computation graphs and backpropagation makes it possible to iterate very quickly on new types of models and loss functions and opens up a new paradigm for model development: **differential programming**.